Too Far to Matter? The Urban Reach of Rural Territorial Conflict

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Can persistent territorial conflict affect nearby urban economies even when conflict events occur outside the city? I study this question in southern Chile, where the Mapuche self-determination conflict is recurrent and predominantly rural. I link geocoded attacks, coercive actions, and land invasions to a panel of 104 fixed urban areas and measure annual external exposure from events occurring within 25 kilometers but outside the receiving area. Greater external conflict exposure is associated with weaker formal business activity in municipal tax records linked to the receiving areas. Moving from the 25th to the 75th percentile of positive annual exposure corresponds to about 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales. The area-clustered 95 percent confidence interval for sales includes zero. Complementary measures of nighttime lights and newly built area show no comparable contemporaneous decline. Fiscal appraisal also does not decline contemporaneously, but turns negative at medium lags. The formal-activity association is not systematically stronger where historical Títulos de Merced cover more surrounding land or where the baseline Mapuche population share is higher. The results show that the economic reach of recurrent territorial conflict can extend beyond contested land and into nearby urban economies, even in a setting where property, tax, Indigenous-land protection, and restitution institutions continue to operate.

Keywords: Indigenous conflict; political violence; local economic activity; spatial spillovers; Chile.
JEL codes: D74; O18; R11.

1 Introduction

Rural conflicts can affect more than the places where violence occurs. Towns and small cities provide markets, services, and economic opportunities for the surrounding areas. When territorial conflict persists in those areas, firms in nearby towns may face disruptions to their activities or reconsider their decisions about investment and location, even when the towns themselves are not directly affected by violence. These consequences may be particularly relevant in regions that already face persistent economic disadvantages. This paper asks whether recurrent territorial conflict is

associated with weaker economic activity in nearby urban areas, and whether this relationship varies with the historical geography of land claims and Indigenous settlement.

A growing literature documents how conflict and insecurity affect firms, investment, and local economic activity. While much of this research focuses on directly exposed locations (Camacho and Rodriguez, 2013; Del Prete et al., 2023), other studies show that the economic consequences of violence can extend beyond the places where it occurs. For instance, Alfano and Cornelissen (2026) show how conflict along transport routes in Somalia affects prices in distant markets, while Abadie and Dermisi (2008) document changes in office vacancy around prominent buildings in Chicago following the September 11 attacks. I examine a different dimension of these spatial spillovers: whether conflict concentrated in rural territories is associated with weaker economic activity in the towns and cities that serve them. I also investigate whether these relationships are reflected in the physical development of urban areas and how they vary across locations within the same city.

Southern Chile provides an interesting setting for studying these questions. Much of the long-standing conflict involving Mapuche territorial claims is concentrated in and around La Araucanía, historically one of Chile's poorest regions. In 1992, its GDP per capita represented approximately 43% of the national average, increasing to 56% by 2017. Despite this partial convergence, La Araucanía ranked last nationally in GDP per capita in 1992, 2002, and 2017.[1] Its persistent economic disadvantage raises questions about the factors that may constrain development in the region, including the potential role of territorial conflict. Larroulet Vignau and Sepúlveda (2025) examine the economic consequences of violence at the regional level. I approach the question from a different spatial perspective, focusing on the economic activity of urban areas exposed to conflict in their surrounding territories.

The Mapuche conflict is particularly interesting because it is closely connected to historical land dispossession, property rights, and the role of the State in addressing territorial claims. Following the incorporation of Mapuche territory into Chile in the late nineteenth century, the State issued *Títulos de Merced* (TDM), establishing a legally recognized geography of Indigenous reservations. Subsequent land transfers, subdivision, and the reversal of agrarian reform substantially changed the distribution of land ownership (Jaimovich and Jordán, 2025a; Jaimovich and Toledo-Concha, 2026). As a result, historical claims and contemporary property rights do not necessarily coincide. Since 1993, the Chilean State has also operated a land restitution programme through the Corporación Nacional de Desarrollo Indígena (CONADI), providing an institutional mechanism for addressing territorial disputes (Gobierno de Chile, 1993).

This historical and institutional context raises several questions about how firms might respond to nearby conflict. One possibility is that territorial disputes increase uncertainty about property rights and future land use, discouraging economic activity in places close to historically recognized claims. Another possibility concerns how firms perceive locations associated with Mapuche communities. If ethnic composition becomes a signal of perceived territorial risk, economic responses

[1]The regional comparisons combine Banco Central GDP and census population data harmonized to the same regional boundaries within each year.

may also vary across places with different historical Mapuche populations, potentially reflecting territorial stigma or statistical discrimination. These interpretations are not mutually exclusive, and ethnic composition may capture other historical and socioeconomic differences. Finally, land restitution introduces the role of public institutions. Resolving historical claims could reduce uncertainty and improve economic expectations, but restitution may also be concentrated in territories where disputes are particularly persistent. Understanding these differences is important for assessing not only the economic reach of conflict, but also the potential role of institutions in addressing its consequences.

To investigate these questions, I combine geocoded conflict events from the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022) with economic and spatial data for 104 fixed urban areas in Biobío, La Araucanía, Los Ríos, and Los Lagos. The main measure captures attacks, coercive actions, and land invasions occurring outside each receiving urban area but within 25 kilometers of its boundary. Conflict occurring inside the receiving area enters separately. I exploit annual variation in nearby conflict exposure, accounting for time-invariant differences across locations, region-specific annual shocks, and trends associated with predetermined characteristics. The analysis combines administrative information on registered enterprises and sales with nighttime lights, fiscal appraisal, cadastral records, and satellite measures of urban development. The tax authority assigns enterprises and sales by municipal tax or headquarters address. I complement the aggregate analysis with a spatial comparison of grid cells within the same urban area and year.

The results show that greater exposure to nearby conflict is associated with weaker formal business activity. Moving from one to ten external events per year, corresponding to the interquartile range among area-years with positive exposure, is associated with approximately 2.2% fewer registered enterprises and 5.3% lower tax-reported sales. The relationship is more precisely estimated for enterprises, while the confidence interval for sales includes zero under the preferred area-clustered inference (Table 1). The dynamic analysis also shows negative enterprise associations over subsequent years, although the timing evidence is less clear for sales. Importantly, these patterns do not extend uniformly to other outcomes. Nighttime lights, fiscal appraisal, and newly built area do not exhibit a common contemporaneous decline. Fiscal appraisal becomes negatively associated with conflict at medium lags, although its administrative revaluation process complicates the interpretation of these adjustments.

The spatial analysis provides a complementary perspective. Within the same urban area and year, grid cells with greater nearby conflict exposure show fewer selected legal entities at short distances and lower shares of commercial cadastral use across several distance thresholds. However, nighttime lights, assessed property values, and new construction do not display a consistent negative spatial gradient (Appendix D). These findings suggest that the relationship between rural conflict and urban economic activity is not uniform across outcomes or locations. They also highlight an important distinction between changes in aggregate formal activity and the distribution of economic activity within cities.

I further examine whether these relationships vary with the historical geography of Indigenous

land claims and population composition. At the municipal level, the negative association with formal business activity is not systematically stronger in places with greater historical TDM coverage in the surrounding area or a larger 1992 urban Mapuche population share (Appendix F). Whether these relationships vary with proximity to historically recognized lands and restitution areas within cities remains open. Recorded CONADI acquisitions under Articles 20(a) and 20(b) are higher two years after greater external conflict exposure (Appendix G). This relationship is consistent with restitution activity responding to persistent disputes, but the available information does not establish whether these interventions subsequently reduce conflict or economic uncertainty.

This paper contributes to the literature by examining an economic dimension of territorial conflict that may be overlooked when analysis focuses exclusively on the locations of violent events or on regional aggregates. The results suggest that the economic consequences of predominantly rural conflict can extend to nearby urban markets, particularly in formal business records, without necessarily producing comparable changes in the physical urban environment. The historical geography of land claims and the institutional response to conflict provide additional perspectives for understanding these relationships, even though the available evidence cannot distinguish conclusively among the proposed mechanisms.

More broadly, the findings speak to the relationship between territorial conflict and regional development. They do not establish that conflict explains La Araucanía's persistent economic disadvantage or that it generates a development trap. Nevertheless, they point to a potentially important economic margin through which recurrent territorial disputes may affect places that are already struggling to catch up. Understanding whether resolving these disputes can improve local economic conditions remains an important question for future research.

2 Institutional Setting and Conflict Geography

2.1 Territorial self-determination and historical claims

The Mapuche are an Indigenous group of south-central Chile whose historical territory encompasses much of the study corridor. Mapuche communities retained substantial autonomy through most of the colonial period and were incorporated into the Chilean State only after the military occupation of Araucanía in the late nineteenth century. Between 1884 and 1929, the State confined Mapuche communities to nearly 3,000 reservations covering roughly 550,000 hectares, a small share of their ancestral territory (Jaimovich and Jordán, 2025b; Jaimovich and Toledo-Concha, 2026). The State formalized these reservations through *Títulos de Merced* (TDM). The TDM therefore did not restore ancestral territory; they legally recognized the much smaller areas left to Mapuche communities after the military occupation. They created an important State-recognized geography of historical territorial claims that remains relevant to later land protection, restitution, and conflict. Importantly, TDM boundaries do not describe current Mapuche ownership. Through subsequent allotment, land sales, encroachment, weak enforcement, and later deregulated transfers, Mapuche communities lost ownership of substantial amounts of land within the historical reserva-

tions (Jaimovich and Jordán, 2025b,a). Contemporary claims can therefore concern land that lies within a historically recognized TDM but is no longer under Mapuche ownership. In the empirical analysis, TDM are used to locate this historical geography of recognized claims, not to identify the current owner of a parcel or the ethnic identity of a nearby firm.

Agrarian reform extended the contested geography beyond the reservations. The reform began in 1962 and accelerated between 1970 and 1973, when Mapuche communities recovered or gained access to land through the expropriation of large estates. After the 1973 military coup, the dictatorship returned or reassigned much of that land, promoted forestry plantations, and divided remaining collective reservations into individual parcels. Contemporary invasions and attacks are more common on plots touched by that reform and its reversal, linking current conflict to grievances both within and beyond TDM boundaries (Jaimovich and Jordán, 2025a; Jaimovich and Toledo-Concha, 2026).

2.2 Formal land institutions and restitution

Chile's 1993 Indigenous Law defines which lands qualify for Indigenous status, including land recognized through *Títulos de Merced*, certain historical occupation, and later State transfers. Articles 13–15 restrict transfers, encumbrances, and use arrangements, with different provisions for community and individually held land. Registration in the Public Registry of Indigenous Lands attests to that legal status (Gobierno de Chile, 1993). In Temuco property records, registered parcels had fewer post-inheritance sales and smaller losses of Mapuche ownership, while registry coverage declined over time (Jordán et al., 2025).

The law also created the Fund for Indigenous Land and Water, administered by CONADI. Article 20(a) subsidizes land acquisition where existing holdings are insufficient. Article 20(b) finances the resolution of land disputes, including purchases related to historically recognized claims (Gobierno de Chile, 1993). The Presidential Commission for Peace and Understanding reports that CONADI acquired 230,563 hectares under Article 20(b) in the four study regions through 2024 (Comisión Presidencial para la Paz y el Entendimiento, 2025, pp. 60–61, 72–73). This is equivalent to about 58 percent of the 396,825 hectares the Commission estimates for its Group 1, which covers applicants with a favorable applicability decision or an application already submitted and allows up to 5 percent growth in beneficiary families. The Commission's broader estimate, including potential future demand, is 486,008 hectares. The ratio compares past purchases with modeled demand for Group 1. It does not measure the fraction of claims already resolved.

Land restitution continued during the study period. Appendix G examines how recorded acquisitions under both articles vary with conflict exposure.

2.3 Conflict geography and recurrence

The Mapuche Conflict Event and Data Archive (MACEDA) records geocoded events from local-media reports in the Mapuche self-determination conflict (Cayul et al., 2022). The main event

family contains attacks, coercive actions, and land invasions. Events recur over time rather than defining a single absorbing transition into conflict.

Most main events occur outside the study's urban areas. Of the 2,775 main events in the four-region corridor, 2,288 (82.5%) lie outside all 104 fixed urban areas. La Araucanía contains 66.6 percent of these events despite accounting for 25.7 percent of the corridor's 2017 population; conflict also occurs in Biobío, Los Ríos, and Los Lagos. Figure 1 maps the events and the locations of the receiving urban areas.

Annual event counts rise and fall throughout the period (Figure 2). Nearby urban areas may therefore face repeated changes in external exposure even when events occur beyond their own urban footprint.

3 Data

3.1 Spatial units and outcomes

The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks within a municipality and remains fixed over time; its footprint can contain spatially disconnected components. These are geographic urban areas, not functional labor-market areas. Their median area is 4.59 km^2 (mean 8.21); the interquartile range is 2.45–7.60 km^2, and the full range is 0.61–63.91 km^2.

Annual records from Chile's national tax authority (Servicio de Impuestos Internos, SII) measure registered enterprises and tax-reported sales. Tax-reported sales are sales declared by firms that SII administratively assigns to the municipality of their tax or headquarters address. Both series are available from 2005 to 2021. These outcomes measure formal activity associated with each urban area; they do not establish that production or sales occur inside the fixed urban footprint.[2]

Nighttime luminosity provides a spatial measure of activity within the fixed urban footprints over a longer period than the SII business records. Lights do not measure formal firms or sales directly, but provide a complementary indicator of broader urban economic activity.

The property cadastre measures total fiscal appraisal and mean appraisal per property within each fixed urban geography. It sums assessed land and, where present, construction values to determine the property-tax base. It is an administratively updated assessed property stock, not a transaction or commercial price. SII updates appraisals through mass revaluations, individual inspections, taxpayer requests, and semiannual CPI indexation. The 2010–2021 panel includes nationally scheduled non-agricultural mass revaluations in 2013, 2014, and 2018; the 2006 round preceded the panel, and the 2013 and 2014 rounds covered different property classes. Annual changes and lagged estimates therefore need not track market-price responses; revaluation schedules, inspections, and indexation govern when assessed values change. Region-by-year effects absorb revaluation-year shocks shared within regions, but not heterogeneous parcel or cadastral changes.[3]

[2] Appendix E examines this spatial interpretation.

[3] See SII descriptions of fiscal appraisal and prior revaluation schedules (Servicio de Impuestos Internos, 2026a,c,b).

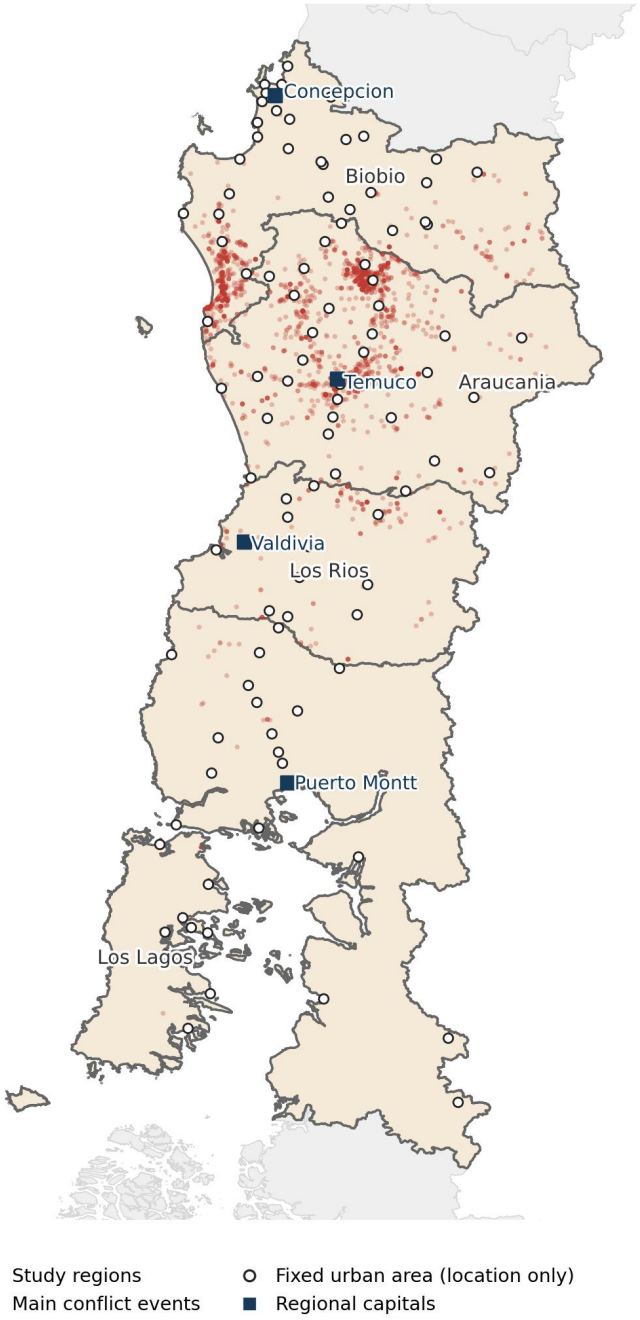


Figure 1: Main conflict events and fixed urban areas in southern Chile. The map shows attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos during 1990–2021. Point markers locate the 104 receiving fixed urban areas, with squares for the four regional capitals; marker size does not represent footprint area.

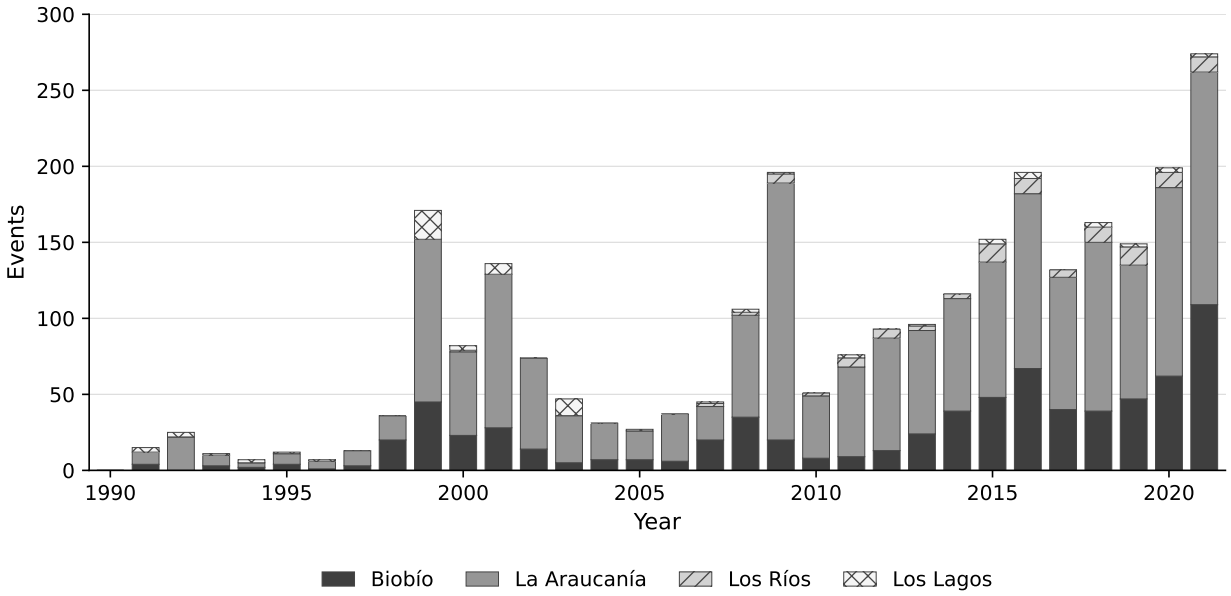


Figure 2: Annual main Mapuche conflict events by region, 1990–2021. Stacked bars count attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos.

WSF Evolution records changes in built surface. It measures a physical flow and complements the economic outcomes by tracking built-area expansion. Its temporal support and interpretation differ from the headline outcomes, so detailed results appear in the appendix. Appendix A documents the construction, transformations, temporal coverage, and spatial support of each series.

3.2 Conflict data and external exposure

Conflict exposure comes from annual geocoded events in the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022). The main event family includes attacks, coercive actions, and land invasions; protests are excluded. Let A_{mt} denote eligible events occurring within 25 kilometers of urban area m in year t, but outside the receiving area. External exposure is

$$D_{mt} = \log(1 + A_{mt}) \,. \tag{1}$$

Conflict counts are zero-heavy and right-skewed. The $\log(1 + x)$ transformation retains zero-exposure area-years and dampens the influence of high counts; coefficients therefore describe changes in transformed exposure and are not elasticities with respect to the raw event count. Appendix B applies the inverse hyperbolic sine consistently to contemporary external conflict, within-area conflict, and cumulative pre-2005 conflict, holding the sample and other controls fixed.

Distance is the exact minimum straight-line distance from each event point to the fixed urban-area footprint, calculated in metric projection EPSG:32718. Events outside the receiving area qualify when this distance is at most 25 km. The exposure includes events outside all study urban areas and events inside another study urban area; it is not restricted to events coded as rural.

Events inside the receiving area enter separately as $I_{mt} = \log(1 + B_{mt})$, where B_{mt} is the annual count of main events inside that area.

The conflict literature does not provide a standardized distance cutoff for nearby exposure. Closely related studies use local windows in the 20–30 km range (Brancati et al., 2026; Fadare et al., 2024), while other work defines exposure along transport corridors (Alfano and Cornelissen, 2026; Liu et al., 2026). I use 25 km based on spatial and connectivity considerations, not outcome estimates, and assess alternative radii from 10 to 60 km. An outcome-free comparison finds that annual area-level exposure counts under the 20–30 km definitions are strongly correlated with counts under a 30-minute contemporary road-network definition (Appendix A.4). The road network uses imputed speeds for most road length and does not reconstruct historical travel over 1990–2021.

Because exposure is defined separately for each receiving area, an event may contribute to multiple areas when their windows overlap. Of 2,775 eligible events in 1990–2021, the preferred mapping contains 8,480 event–area pairs. Each event enters about 3.1 areas on average (median 3; maximum 9), while 109 events (3.9 percent) enter no area's external exposure. Events outside all study urban areas make up 81.7 percent of receiver–event pairs; events inside another study urban area make up the remainder. Each area-year remains one panel observation. Figure A.3 shows the event-level distribution. Both location categories contribute to the preferred exposure measure.

4 Empirical Strategy

4.1 Baseline specification and identifying variation

For outcome Y_{mt}, the preferred area-level specification is

$$Y_{mt} = \beta D_{mt} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} + \tau_t (X_m^0)'\theta + \varepsilon_{mt}, \tag{2}$$

where X_m^0 contains standardized 2002 municipal log population, 2002 municipal rural share, and cumulative exact-25-km external conflict from 1990 through 2004. The terms μ_m and $\lambda_{r(m),t}$ are area and region-by-year fixed effects, and τ_t is linear calendar time. Standard errors are clustered by area. Because nearby areas can share exposure and local shocks, Appendix B also reports spatial-HAC inference allowing spatial and temporal residual dependence.

The preferred specification allows secular trajectories to differ with initial population, initial rural composition, and pre-sample external conflict exposure. Population and rural share are measured for the municipality in the 2002 Census, not within the later fixed urban footprint. Population enters as $\log(1 + P_m^{2002})$. Historical conflict uses the same broad-external exact-25-km definition as contemporary exposure, summing eligible events from 1990 through 2004 and then taking $\log(1 + H_m)$. These interactions allow areas with different initial scale, rural composition, and prior conflict geography to follow different linear trajectories. Appendix B documents their construction.

Region-by-year effects absorb annual shocks shared by areas within Biobío, Araucanía, Los Ríos,

and Los Lagos, including regional economic and policy conditions. The model also conditions on contemporaneous conflict inside the receiving area. The external coefficient therefore captures the association with nearby conflict conditional on direct local conflict; this co-exposure is not assumed to be exogenous.

The characteristics predate the 2005–2021 formal-activity and 2010–2021 appraisal panels. They do not predate the full lights and built-area windows, beginning in 1992 and 1990. Those longer-window models retain area and region-by-year effects and within-area conflict but omit all three trends.

Unobserved local shocks could affect both conflict and economic activity. Area effects remove persistent differences, region-by-year effects remove shared regional shocks, and the trends allow trajectories to differ with initial conditions. The coefficient β uses the remaining within-area variation in exposure. It is a conditional association because this adjustment does not make subsequent conflict random.

4.2 Timing patterns

The timing analysis extends the preferred specification by including future, contemporaneous, and lagged values of the same annual exposure measure:

$$Y_{mt} = \sum_{h=1}^{3} \pi_h D_{m,t+h} + \sum_{\ell=0}^{L_Y} \beta_\ell D_{m,t-\ell} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} + \tau_t (X_m^0)'\theta + \varepsilon_{mt}. \tag{3}$$

The preferred projection includes three future-exposure terms and lags through eight years for enterprises, sales, and fiscal appraisal. The nighttime-lights profile extends through ten years because of its longer temporal support.

For presentation, coefficients on future exposure are plotted at negative horizons. These terms show whether outcomes tend to move before the corresponding exposure is realized, while the contemporaneous and lagged coefficients trace how the association evolves during and after exposure. Because conflict exposure is continuous and recurrent, the exercise is best read as a description of its temporal pattern rather than as a conventional event study.

Appendix C.2 examines dynamic adjustment around extraordinary conflict escalations using first-exposure C&S (Callaway and Sant'Anna, 2021), repeated-exposure DiD-style comparisons, and native nonabsorbing DCDH (de Chaisemartin and D'Haultfœuille, 2026). These exercises compare different exposure histories and populations and do not inherit the main specification's full covariate-trend adjustment. They provide complementary timing evidence, not an independent causal identification strategy.

5 Results

5.1 Baseline estimates

Nearby external conflict is associated with lower formal business activity in municipal tax records linked to the receiving urban areas. In the preferred specification, moving from the 25th to the 75th percentile of positive exposure corresponds to about 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales. The area-clustered 95 percent confidence interval for sales includes zero (Table 1).

Table 1: External conflict exposure and urban-area outcomes

Specification	(1) Registered enterprises	(2) Tax-reported sales	(3) Nighttime lights	(4) Fiscal appraisal total	(5) WSF new built area
Area + year FE	-0.020**	-0.050**	-0.005	0.001	-0.041
	(0.009)	(0.022)	(0.011)	(0.021)	(0.034)
+ predetermined trends	-0.012**	-0.034**	—	0.002	—
	(0.006)	(0.017)		(0.016)	
+ region-by-year FE	-0.013**	-0.032*	-0.004	0.019	0.018
	(0.007)	(0.019)	(0.012)	(0.013)	(0.031)
+ within-area conflict (preferred)	-0.013**	-0.032*	-0.004	0.019	0.021
	(0.006)	(0.019)	(0.012)	(0.013)	(0.030)
Observations	1,768	1,768	3,060	1,248	2,704
Areas	104	104	104	104	104
Sample period	2005–2021	2005–2021	1992–2021	2010–2021	1990–2015

Notes: External exposure counts attacks, coercive actions, and land invasions within 25 km of the fixed urban footprint, excluding events inside the receiving area. All specifications include area fixed effects. The four rows add year effects, the three predetermined trends, region-by-year effects, and contemporaneous within-area conflict. The trends interact calendar time with 2002 municipal population, 2002 rural share, and cumulative external conflict in 1990–2004. Predetermined trends are not estimated for nighttime lights and WSF because their panels begin before the baseline characteristics. WSF is measured in asinh-transformed hectares. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The enterprise and sales associations remain negative as the specification adds predetermined trends, region-by-year effects, and within-area conflict. The trends allow outcome paths to differ with initial population, rural composition, and prior conflict. The contemporaneous estimates for lights, appraisal total, and new built area show no common contraction: lights are close to zero, while appraisal and built area have positive, imprecise estimates.

The enterprise and sales estimates remain negative across exposure radii from 10 to 60 km, including 20, 25, and 30 km (Appendix B). These specifications vary the current exposure radius while holding the historical trend at 25 km.

The negative enterprise association is not confined to firms classified in forestry or wood and paper manufacturing. Excluding those activities changes the coefficient from -0.0132 to -0.0127, with wider uncertainty (Table B.6). Retained firms may still depend on forestry demand.

5.2 Timing patterns

The continuous timing profile shows a persistent negative enterprise association over the first several years after exposure, with the strongest pattern around the medium-term lags (Figure 3). Future exposure does not show a comparable systematic enterprise pattern. That absence does not show that conflict is unanticipated or exogenous. Because exposure recurs and remains continuous, these lead-lag estimates describe timing rather than isolate a one-time treatment effect.

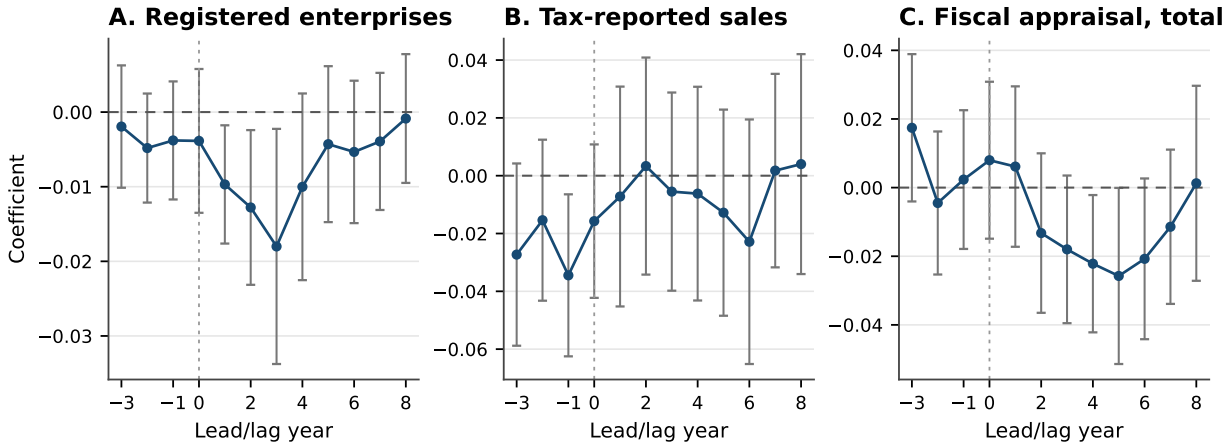


Figure 3: Timing patterns in formal activity and fiscal appraisal. The panels report coefficients from the continuous lead-lag projection defined in Section 4. Negative values on the horizontal axis are future exposure terms, zero is current exposure, and positive values are lagged exposure terms. Vertical bars are 95 percent confidence intervals based on standard errors clustered by fixed urban area. All specifications use broad external exposure within 25 kilometers, condition on contemporaneous within-area conflict, and include area and region-by-year effects and the three predetermined trends defined in Section 4.1.

Sales estimates are generally negative, but their uncertainty leaves no clear dynamic profile. Fiscal appraisal does not fall contemporaneously; its negative pattern emerges at medium lags. Because appraisal records an administratively updated property stock, that delay may reflect when values are reassessed, and does not by itself establish a delayed causal effect.

The supplementary escalation comparisons distinguish first experience from repeated exposure and subsequent treatment switches (Appendix C.2). Their different populations, selected comparison paths and adverse nighttime-lights preperiods limit a causal interpretation. The continuous profile remains the main timing evidence; the escalation comparisons do not identify the causal cost of persistent conflict.

5.3 Spatial interpretation and supporting evidence

SII outcomes are assigned through municipal tax and headquarters addresses, so they do not by themselves locate all reported activity inside urban areas. For the selected legal entities that can be matched to cadastral headquarters, the negative association is also visible among urban-

headquartered entities (Appendix E.1).

The grid comparison asks whether urban cells with greater external exposure also show weaker local economic activity or land use than other cells in the same area and year. More exposed cells have fewer selected legal entities at 5 and 10 km, with no clear negative association at 25 km. Commercial-use shares are lower at 10, 20, 25, and 30 km, while lights, appraisal, and newly built area show no common gradient (Appendix D). These within-area differences do not measure aggregate urban decline or establish that activity moved between cells.

If the municipal SII association mainly reflects firms tied to the surrounding rural economy, sectors with stronger predetermined agricultural or forestry input and customer linkages should respond more strongly. The estimates show no systematic gradient by these linkages (Appendix E).

6 Additional Evidence

6.1 Robustness

The enterprise and sales estimates remain negative as the model moves from common year effects to region-by-year and province-by-year effects, although the finer geographic controls absorb more exposure variation and reduce precision (Appendix B). Applying the asinh transformation consistently to conflict measures produces similar enterprise and sales magnitudes for the same change in raw exposure.

Excluding aquaculture, the sector most directly exposed to the 2007–2010 Los Lagos crisis (Ritter, 2025), leaves the enterprise estimate almost unchanged. In the common area-years with subsector-sales data, the sales estimate remains negative but imprecise. Alternative wild-cluster and spatial-HAC inference supports the same negative formal-activity pattern, though uncertainty varies with the assumed dependence across areas. These procedures do not address confounding.

The formal-activity association remains similar after controlling for nearby protests and protests inside the receiving area. This makes it less likely that the baseline estimates simply reflect broader episodes of Mapuche-related mobilization (Appendix B.2, Table B.2).

The estimates do not show a systematically stronger association in areas with greater historical *Títulos de Merced* coverage or a larger 1992 Mapuche population share (Appendix F).

6.2 Margins of adjustment

The enterprise stock is more negatively associated with nearby conflict at transport locations than elsewhere. The series SII labels first registrations has a similar pattern, but its association is sensitive to the observation window and SII publishes identical data under two distinct registration labels. Recorded terminations show no clear association before a separate administrative break in 2016–2017. These series cannot separate the contribution of firm entry and exit to the change in the enterprise stock (Appendices B.8 and B.9).

7 Discussion and Conclusion

This paper studies whether persistent territorial conflict is associated with weaker business activity in nearby urban areas. In southern Chile, moving from the 25th to the 75th percentile of positive external exposure corresponds to 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales in municipal records linked to the receiving areas. The area-clustered confidence interval for sales includes zero. The comparison accounts for regional annual shocks, pre-sample characteristics, and contemporaneous conflict inside the receiving area.

Nighttime lights, fiscal appraisal, and newly built area show no common contemporaneous decline. Appraisal coefficients turn negative at medium lags in the continuous projection, which may partly reflect reassessment timing. Within the same urban area and year, the association with selected legal entities is negative at 5 and 10 km and imprecise at 25 km. Commercial-use shares are lower at several radii. The negative association also appears among selected entities headquartered in urban areas. Municipal tax addresses leave the location of some reported sales and production unresolved.

The municipal formal-business association does not systematically vary with historical title coverage or the 1992 urban Mapuche population share. The more negative enterprise association for transport-location conflict is consistent with disruptions to access, although security concerns and expectations about surrounding land may also matter. Greater conflict exposure precedes more recorded CONADI acquisition activity at the two-year lag. The available dates and pooled program records leave open whether restitution changes subsequent conflict or economic activity.

The clearest evidence of economic reach beyond the places where conflict occurs appears in the formal business records of nearby municipalities. That association can coexist with functioning property, tax, and restitution institutions, and with little contemporaneous change in the built environment.

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A Data, measurement, and sample construction

A.1 Fixed urban-area geography and study sample

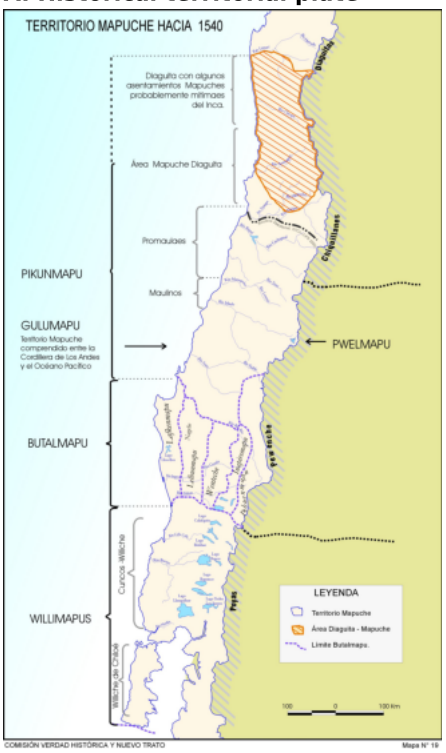
The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks ("manzanas" with `AREA=1`) within a municipality and is held fixed over time; its footprint can contain spatially disconnected components. The four regions cover the part of south-central Chile where historical Mapuche occupation, recognized claims, contemporary restitution, and recent conflict most clearly overlap. The sample is selected for this overlap and is not nationally representative.

SII enterprise and sales records are observed at municipal tax addresses and mapped to the corresponding urban area; they therefore measure formal activity associated with the area rather than activity mechanically located inside its urban footprint.

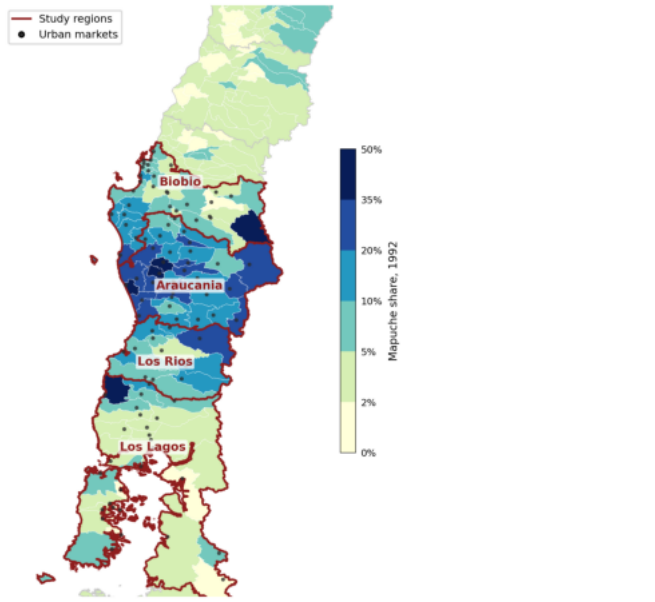
Figure A.1 provides historical and demographic context for the study corridor.

The business panel covers 2005–2021. Fiscal appraisal begins in 2010, while harmonized nighttime lights and WSF built-area data have longer, but distinct, temporal support. Table A.2 records the construction and transformation of each outcome used in the analysis.

A. Historical territorial plate



B. Contemporary corridor and 1992 Mapuche shares



Darker shades mark communes with larger Mapuche population shares in 1992; points show the urban markets used in the paper.

Figure A.1: Historical Mapuche territory and the contemporary study corridor. Panel A reproduces a historical map of Mapuche territorial organization around 1540 from the Comisión Verdad Histórica y Nuevo Trato. Panel B shows the four-region study corridor and Mapuche population shares in 1992. Together, the panels illustrate the historical and demographic geography underlying the study area.

A.2 Conflict data and event composition

Conflict events come from MACEDA. The main analysis counts attacks, State coercive actions, and land invasions; protests are excluded. This family contains 2,775 events, of which 82.1 percent have MACEDA's rural classification and 82.5 percent lie outside all fixed urban areas used to construct the exposure measure. Most events are therefore rural, but an event qualifies as external exposure when it occurs outside the receiving urban area.

Table A.1: MACEDA event composition and regional concentration

Event family	Events	Rural MACEDA	Share	
Panel A. Event composition				
All MACEDA events	3,730	2,587	69.4%	
Main event family: attacks, coercion, land invasions	2,775	2,278	82.1%	
Protests	955	309	32.4%	
Panel B. Operational fixed-system geography of the main event family				
Classification	Events	Share		
Outside all fixed urban systems	2,288	82.5%		
Inside a fixed urban system	487	17.5%		
Panel C. Regional concentration of main events				
Region	Main events	Event share	2017 population	Population share
Biobio	752	27.1%	1,556,805	41.8%
La Araucanía	1,847	66.6%	957,224	25.7%
Los Lagos	76	2.7%	828,708	22.2%
Los Ríos	100	3.6%	384,837	10.3%
Panel D. La Araucanía share of cumulative main events				
Window	Main events	La Araucanía events	Share	
1990–1992	40	30	75.0%	
1990–2002	589	401	68.1%	
1990–2017	1,990	1,371	68.9%	
1990–2021	2,775	1,847	66.6%	

Notes: The main event family comprises attacks, coercive actions, and land invasions. Rural is MACEDA's event classification. The outside-area classification uses the fixed urban geography. Protests are not used in the exposure measure.

A.3 External exposure construction

Figure A.2 illustrates the receiving-area definition in Section 4.1, including how events inside one urban area can contribute to another area's external exposure. Figure A.3 shows how many receiving areas each event enters.

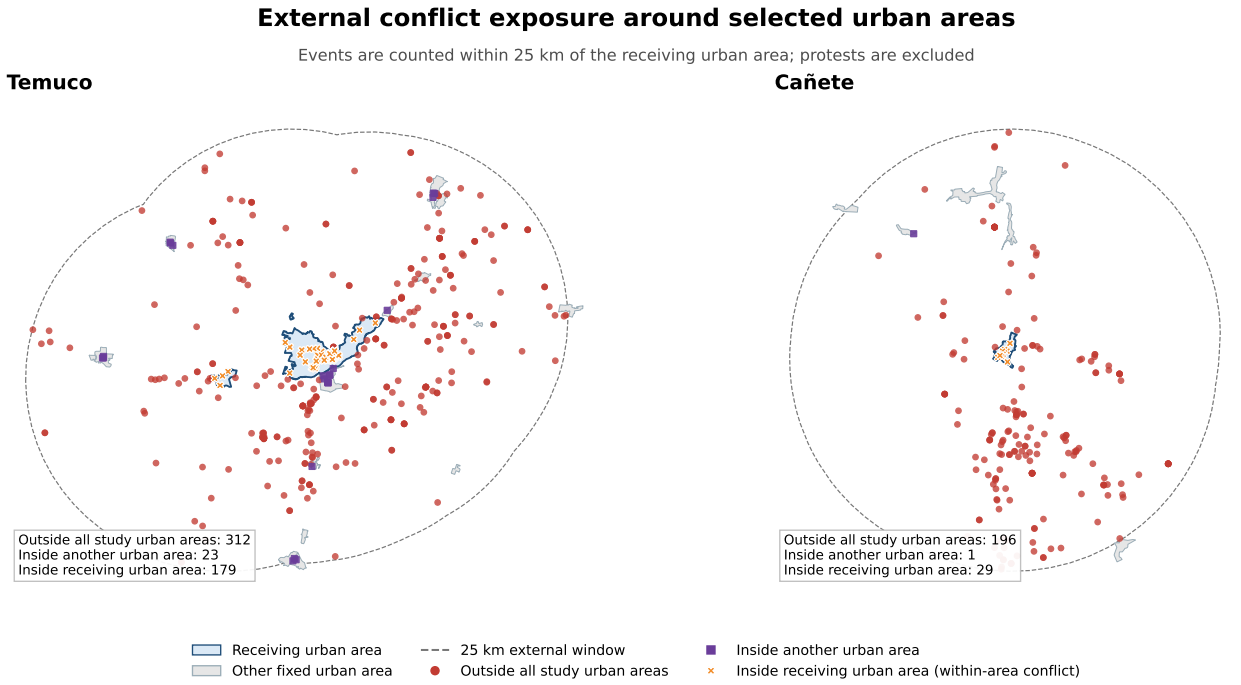


Figure A.2: External conflict exposure around selected urban areas. The dashed boundary marks the 25-kilometer window around each receiving area. Events outside all study urban areas and events inside another study urban area contribute to external exposure. Events inside the receiving area are shown separately as within-area conflict.

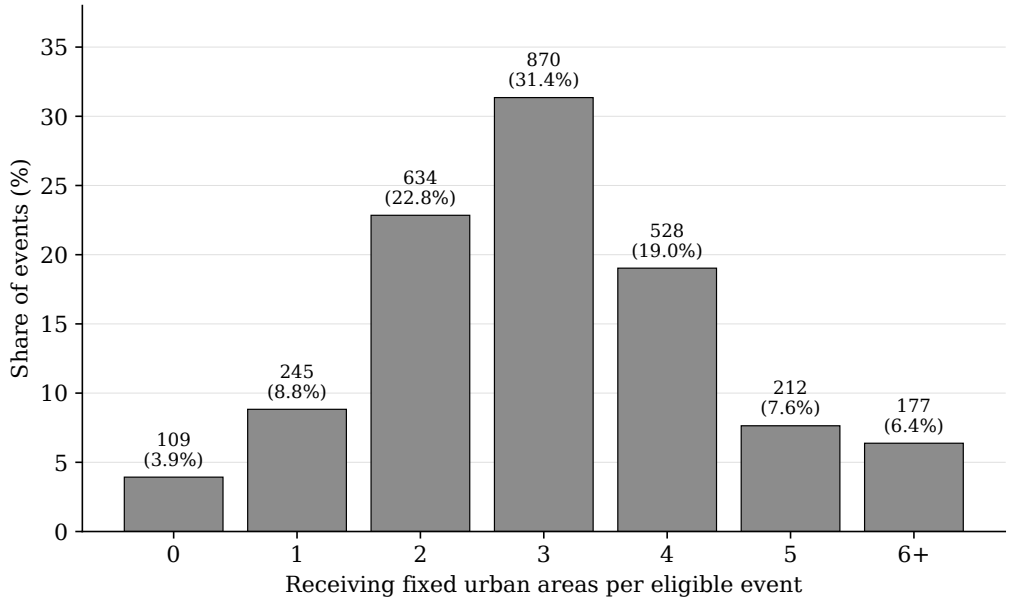


Figure A.3: Event multiplicity under exact 25 km external exposure. Bars show the share of 2,775 eligible MACEDA events entering the exposure of 0, 1, 2, 3, 4, 5, or at least 6 receiving fixed urban areas. Labels report counts and shares. Own-area events are excluded from that area's external count, but may enter another area's exposure.

A.4 Geographic distance and road accessibility

This outcome-free diagnostic compares exact event-point-to-footprint distance with fastest-route travel time on a contemporary road network. The September 2026 OpenStreetMap routing graph contains 122,772 km of motor-vehicle roads; numeric posted speeds cover 14.99 percent of road length in this routing crop, and class-based speeds are assigned to the remaining 85.01 percent. All 188 components of the 104 urban footprints serve as origins. Events are snapped to the nearest road segment, with a 2-km maximum; 22 events fail this cutoff or fall outside the routing crop and are omitted from road-time assignments. Ferries are excluded.

Definition	Event–area pairs	Jaccard vs. road	Annual-count corr.	Mismatch (% / %)*
Euclidean 20 km	6,614	0.576	0.905	38.0 / 10.8%
Euclidean 25 km	8,480	0.628	0.925	27.1 / 18.2%
Euclidean 30 km	11,455	0.640	0.922	13.9 / 28.5%
OSM road 30 min	9,513	—	—	—

*Both mismatch quantities are percentages: road pairs missed by the Euclidean definition / Euclidean pairs excluded by the road definition. Annual-count correlations use area-year counts for 1990–2021.

The annual counts correlate most with the 25-km definition, while pair-level Jaccard agreement is highest for 30 km. Neither comparison identifies an optimal radius, and Euclidean distance is not equivalent to driving time. These assignments describe spatial connectivity only: the network postdates the 1990–2021 event panel, speeds are partly imputed, and historical travel times are not reconstructed. The diagnostic contains no outcomes and tests no economic mechanism.

A.5 Outcome construction and SII measurement

Nighttime lights, fiscal appraisal, and WSF are extracted from fixed urban footprints. Fiscal appraisal is an assessed administrative stock, not a transaction price, and is updated more slowly than a high-frequency market price. WSF records newly built area and therefore measures a physical flow. For registered enterprises and tax-reported sales, the SII address is a tax or headquarters address; it does not identify where a firm produces or where its sales take place. Appendix E examines the implication of this distinction.

A.6 Exposure support

Table A.3 describes the support for external and within-area exposure in the full 1990–2021 area panel. Within-area conflict captures a distinct contemporaneous geography from the external-exposure measure.

Table A.2: Outcome construction and interpretation

Outcome	Source and spatial unit	Years	Transformation	Interpretation
Registered enterprises	SII municipality tax-address measure mapped to area	2005–2021	log(enterprises)	Formal enterprise stock associated with the area
Tax-reported sales	SII municipality tax-address measure mapped to area	2005–2021	log(sales)	Scale of formal activity associated with the area
Nighttime lights	Harmonized DMSP–VIIRS, fixed urban footprint	1992–2021	log(total DN)	Urban luminosity
Fiscal appraisal total	SII cadastre, fixed urban footprint	2010–2021	log(appraised value)	Assessed urban property stock
Fiscal appraisal mean	SII cadastre, fixed urban footprint	2010–2021	log(mean appraisal)	Mean assessed property value
WSF new built area	WSF Evolution, fixed urban footprint	1990–2015	asinh(new hectares)	New built-area flow

Notes: Registered enterprises and tax-reported sales are SII municipality tax-address measures mapped to urban areas. Nighttime lights, fiscal appraisal, and WSF are constructed within fixed urban footprints. Transformations follow each outcome's support and unit; they are not all logs.

Table A.3: Support for external and within-area conflict exposure

Exposure measure	Mean	SD	Positive area-years
External exposure, D_{mt}	0.556	0.947	33.6%
Within-area exposure, I_{mt}	0.063	0.283	5.9%
$\text{Corr}(D_{mt}, I_{mt}) = 0.374$; both positive: 4.7%			

Notes: $D_{mt} = \log(1 + A_{mt})$ is broad external exposure; I_{mt} is the log of one plus annual main conflict events inside the receiving area. The panel contains 104 areas from 1990 to 2021.

B Robustness and margins of adjustment

B.1 Where external conflict occurs

External exposure can include events outside every study urban area or events inside another study urban area but outside the receiving city. Events in another city still contribute to the receiving area's nearby exposure, so the comparison asks whether those events account for the main association. About 81.7 percent of external event-area pairs originate outside all study urban areas. Table B.1 reports these shares and enters the two locations jointly in the preferred specification. The enterprise association remains negative for events outside all study cities. For sales, both coefficients are negative, but the outside-city estimate is imprecise while the inside-another-city estimate is more precise. The equality tests do not distinguish the two coefficients, but they do not establish that the components have equal effects. The evidence therefore weighs against an other-city-only explanation for enterprises, while leaving the location contribution to sales unresolved.

Table B.1: Where external conflict occurs

Panel A. Exposure composition, 1990–2021	Outside all study urban areas	Inside another study urban area	All external events
Event–area pairs	6,931	1,549	8,480
Share of pairs	81.7%	18.3%	100.0%
Area-years with positive exposure	28.2%	14.6%	33.6%
Panel B. Joint exposure estimates, 2005–2021	Outside all study urban areas	Inside another study urban area	Equality p-value
Registered enterprises	-0.0139*	-0.0029	
	(0.0074)	(0.0062)	0.242
Tax-reported sales	-0.0162	-0.0498**	
	(0.0218)	(0.0225)	0.275

Notes: Eligible events are attacks, coercive actions, and land invasions outside the receiving area and within 25 km. Panel A counts event–area pairs in 1990–2021 and reports the share of area-years with positive exposure. The two component exposures have a correlation of 0.449. Panel B enters both locations jointly in 2005–2021 with area and region-by-year effects, the three predetermined trends, and contemporaneous within-area conflict. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Conflict composition and protest exposure

Exposure to attacks, coercive actions, and land invasions is strongly correlated with nearby protest exposure (0.718 across area-years in 2005–2021). Yet controlling jointly for external protests and protests inside the receiving area leaves the enterprise and sales coefficients on main-family exposure close to their baseline estimates (Table B.2). Protest coefficients are imprecise; adding a pre-2005 protest-history trend makes the enterprise protest estimate more negative without materially changing the main-family estimates. The coefficient comparisons do not establish different associations for conflict and protests.

MACEDA codes attacks and land invasions overwhelmingly to Mapuche actors and coercive actions overwhelmingly to the State. Among State-coded coercive actions in 2005–2021, Carabineros account for 361 of 487 events, followed by the judiciary (91) and PDI (27). Coercive actions include raids, repression, arrests, evictions, convictions, indictments, and coercive threats. When the source does not identify the initiator, MACEDA defaults to ACTOR1; these coded shares describe event-family composition, not verified initiation in every case.

In the joint two-group model, coercive-action estimates are negative for both formal outcomes, especially sales; the enterprise interval still permits a modest positive association. The contentious-action and coercive-action coefficients are not statistically distinguished. State responses may occur where underlying conflict is more severe, so these estimates do not identify the effect of State intervention.

Table B.2: Conflict composition, protest exposure, and formal activity

Exposure / specification	Registered enterprises	Tax-reported sales
Panel A. Main conflict and protest exposure		
Baseline: main-family exposure	-0.0132**	-0.0320*
	(0.0065)	(0.0189)
Protest-only exposure	-0.0073	-0.0012
	(0.0061)	(0.0199)
Joint: main-family exposure	-0.0126*	-0.0362*
	(0.0065)	(0.0195)
Joint: protest exposure	-0.0026	0.0124
	(0.0059)	(0.0202)
Joint + protest-history trend: main family	-0.0143**	-0.0374*
	(0.0065)	(0.0193)
Joint + protest-history trend: protests	-0.0094	0.0073
	(0.0061)	(0.0201)
Panel B. Contentious and coercive actions		
Contentious actions	-0.0140*	-0.0103
	(0.0077)	(0.0246)
Coercive actions	-0.0033	-0.0424**
	(0.0065)	(0.0208)
Equality of exposure coefficients (p)	0.296	0.393
Area-years	1,768	1,768
Urban areas	104	104

Notes: The sample covers 104 fixed urban areas in 2005–2021 (1,768 area-years). Outcomes and exposures are in logs; event counts use $\log(1+x)$. External events occur outside the receiving area and within exact 25-km point-to-footprint distance. All models include area and region-by-year effects and the three preferred predetermined trends, with standard errors clustered by area. The baseline controls for contemporaneous within-area main-family conflict. Protest-only and joint protest models control for both contemporaneous within-area main-family conflict and within-area protests. The additional protest-history trend uses pre-2005 external protests. Panel B enters external contentious and coercive actions jointly with their matching within-area exposures; contentious actions sum attack and land-invasion raw counts before $\log(1+x)$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Transformation and inference sensitivity

The preferred model includes linear trends based on 2002 municipal population and rural share and cumulative external conflict in 1990–2004 at exact 25 km.

Table B.3 replaces $\log(1+x)$ with asinh for contemporary external and within-area counts and the cumulative pre-2005 count before its trend interaction. Sample and other controls are fixed. A common increase from one to ten events gives similar enterprise and sales associations; raw coefficients are not directly comparable.

Table B.3: Comparable raw-count contrasts under consistent conflict transformations

Outcome	Transformation	1 to 10 events (%)	95% CI	p
Registered enterprises	$\text{Log}(1+x)$	-2.22	[-4.34, -0.06]	0.045
Registered enterprises	Consistent asinh	-2.13	[-4.12, -0.10]	0.040
Tax-reported sales	$\text{Log}(1+x)$	-5.30	[-11.17, 0.95]	0.094
Tax-reported sales	Consistent asinh	-5.53	[-11.19, 0.50]	0.071

Notes: Percent associations equal $100\{\exp[\hat{\beta}(g(10)-g(1))]-1\}$, with g equal to $\log(1+x)$ or asinh. Outcomes are in logs. All three conflict quantities use the stated transformation. Both models use 1,768 area-years and area-clustered intervals.

Wild-cluster inference gives $p = 0.041$ for enterprises and $p = 0.079$ for sales (Table B.4). At the 60 km spatial cutoff and three annual lags, spatial-HAC gives $p = 0.013$ and $p = 0.019$. The 30 and 90 km cutoffs give similar dependence-adjusted precision. HAC uses a normal reference distribution; area-clustered headline intervals use t with 103 degrees of freedom. Neither addresses endogeneity.

Table B.4: Alternative inference for headline formal outcomes

Outcome	Coef.	Clustered SE	Wild p	Spatial-HAC SE	Spatial-HAC p
Registered enterprises	-0.0132	(0.0065)	0.041	(0.0053)	0.013
Tax-reported sales	-0.0320	(0.0189)	0.079	(0.0136)	0.019

Notes: Wild-cluster p-values use 999 repetitions and area clustering. Spatial-HAC standard errors use a 60-kilometer spatial cutoff and a three-year lag cutoff; 30- and 90-kilometer cutoffs give the same qualitative conclusion. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Province influence

Omitting each province in turn leaves the enterprise and sales estimates negative, although precision varies with the retained sample (Figure B.1).

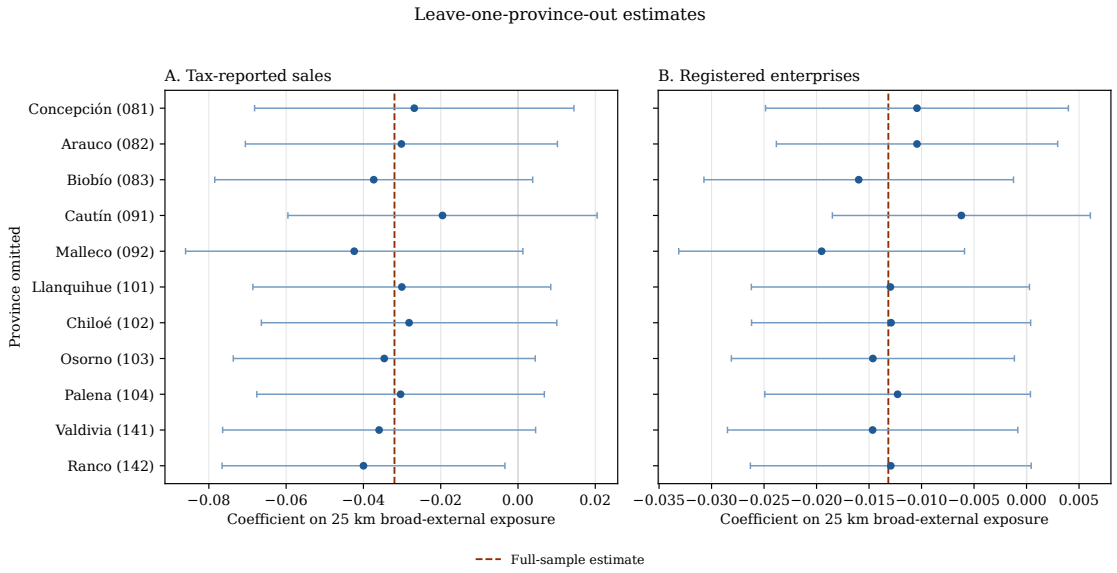


Figure B.1: Leave-one-province-out estimates for formal activity. Each point reports the estimate obtained after omitting the indicated province. Vertical lines show the corresponding full-sample estimates, and horizontal bars report 95 percent confidence intervals.

B.5 Distance sensitivity

The main specification measures external conflict within 25 kilometers of each urban area. To assess whether the formal-activity results depend on this particular spatial window, Table B.5 reconstructs external exposure using cumulative radii from 10 to 60 kilometers. The coefficients remain negative for both registered enterprises and tax-reported sales throughout the range. I interpret this exercise as sensitivity to the spatial window rather than as evidence for an optimal radius of influence.

Table B.5: Cumulative external-conflict windows and formal activity

External window	Registered enterprises	Tax-reported sales
Within 10 km	-0.0133* (0.0078)	-0.0209 (0.0245)
Within 20 km	-0.0138** (0.0055)	-0.0292 (0.0195)
Within 25 km (preferred)	-0.0132** (0.0065)	-0.0320* (0.0189)
Within 30 km	-0.0181*** (0.0064)	-0.0478*** (0.0163)
Within 40 km	-0.0172*** (0.0062)	-0.0456** (0.0225)
Within 50 km	-0.0170** (0.0070)	-0.0414* (0.0238)
Within 60 km	-0.0166** (0.0079)	-0.0451* (0.0246)

Notes: Each row replaces the preferred 25-kilometer exposure measure with log external conflict accumulated within the indicated radius. The event family, within-area conflict, area and region-by-year effects, demographic trends, and area clustering are unchanged. History uses exact-25-km broad-external events in 1990–2004 in every row; only current external exposure varies. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6 Forestry exclusions

The negative enterprise association persists after removing firms assigned to forestry activities (Table B.6). Narrow forestry excludes forest nurseries and silviculture, logging, nonwood forest collection, and forestry services (SII groups 021–024). Broad forestry also excludes sawmilling, wood and related products, and pulp and paper manufacturing (groups 161, 162, and 170). The table reports both definitions separately and then excludes aquaculture (group 032) alongside each.

Enterprise counts reconcile with the published municipal totals. All five enterprise estimates use 104 areas over 2005–2021. The coefficient changes from −0.0132 for all enterprises to −0.0122 without narrow forestry and −0.0127 without broad forestry. Joint exclusion of aquaculture leaves both coefficients close to their forestry-only values. The 95 percent confidence intervals for the exclusion estimates include zero.

Panels B and C report a separate sensitivity that assigns zero to sales cells that SII suppresses with an asterisk. Reported numeric zeros remain zeros, other missing entries remain missing, and suppressed values are unknown. The resulting sales measures are approximate. Subtracting only published sales can leave undisclosed sales from excluded sectors in the official total. Summing retained subsectors can omit undisclosed sales in those subsectors. The exclusion estimates are negative under both constructions, but neither isolates actual sales outside forestry.

B.7 Aquaculture exclusion

Los Lagos contains the largest concentration of aquaculture activity in the study corridor. Because this sector represents a large share of observed sales in the region, a sector-specific downturn could potentially contribute to the aggregate formal-activity relationship. I therefore reconstruct both SII outcomes after excluding subsector 032 (Acuicultura).

Excluding aquaculture leaves all 104 areas and enterprises almost unchanged: −0.013162 becomes −0.013152. Suppression of subsector sales restricts the comparison to 1,066 common area-years. Total sales in that sample gives −0.0124 (SE 0.0216), compared with −0.0144 (SE 0.0216) without aquaculture. Both estimates are imprecise. This is a comparison among area-years with published numeric subsector sales and positive support for both logged totals. Disclosure can select area-years with different aquaculture activity, so the paired comparison does not describe the full panel.

B.8 Margins of adjustment

Recorded terminations count formal cessations that remain current at the date of extraction. Taxpayers that subsequently restarted are excluded. This retrospective administrative measure covers the taxpayer universe rather than only the enterprises in the stock series. Both terminations and the SII series published as first registrations enter as $\log(1 + x)$, without reversing their signs. A negative termination coefficient therefore means fewer recorded terminations, not an increase in exits.

Table B.6: Formal activity excluding forestry and related manufacturing

Outcome definition	Coefficient	SE	p-value	Area-years
Panel A. Registered enterprises				
All activity	-0.0132	0.0065	0.045	1,768
Excluding narrow forestry	-0.0122	0.0066	0.067	1,768
Excluding broad forestry	-0.0127	0.0067	0.062	1,768
Excluding narrow forestry and aquaculture	-0.0122	0.0066	0.068	1,768
Excluding broad forestry and aquaculture	-0.0127	0.0067	0.063	1,768
Panel B. Official sales less excluded subsector sales (zero imputation)				
All activity	-0.0320	0.0189	0.094	1,768
Excluding narrow forestry	-0.0219	0.0194	0.263	1,768
Excluding broad forestry	-0.0233	0.0195	0.235	1,768
Excluding narrow forestry and aquaculture	-0.0226	0.0195	0.249	1,768
Excluding broad forestry and aquaculture	-0.0240	0.0196	0.223	1,768
Panel C. Subsector sums with suppressed cells set to zero				
All activity	-0.0508	0.0222	0.024	1,765
Excluding narrow forestry	-0.0425	0.0243	0.083	1,765
Excluding broad forestry	-0.0461	0.0248	0.066	1,765
Excluding narrow forestry and aquaculture	-0.0411	0.0243	0.094	1,765
Excluding broad forestry and aquaculture	-0.0446	0.0247	0.074	1,765

Notes: Narrow forestry excludes SII groups 021–024; broad forestry also excludes 161, 162, and 170. Group 162 includes a mixed category of wood, cork, straw, and plaiting products. Aquaculture is group 032. Enterprise counts sum firms by assigned activity. Sales in Panel B equal the official municipal total in UF less the published sales of excluded subsectors; Panel C sums sales of retained subsectors. For these sales constructions, literal published ∗ cells are set to zero while numeric cells are unchanged. Table B.7 uses published numeric sales instead. Outcomes are logged on positive support. Three zero constructed sums leave Panel C with 1,765 area-years. Models use the preferred specification in Section 4.1 on 104 areas in 2005–2021, with area-clustered standard errors.

Table B.7: Formal activity excluding aquaculture

Outcome	Main estimate	Excluding subsector 032	Area-years
Registered enterprises	-0.0132** (0.0065)	-0.0132** (0.0065)	1,768
Tax-reported sales	-0.0124 (0.0216)	-0.0144 (0.0216)	1,066

Notes: Aquaculture corresponds to SII subsector 032. The enterprise comparison uses the complete 104-area panel. Because subsector-level sales are suppressed in some municipality-years, the sales comparison uses the same disclosed, positive-support sample before and after excluding aquaculture. All specifications otherwise reproduce the main area-level model. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The reference termination window is 2005–2015. SII documents administrative actions in 2016 and 2017 that increased recorded terminations nationally. Before those actions, the coefficient is −0.0116 (SE 0.0218; $p = 0.595$; 1,144 area-years), with a 95 percent interval of $[-0.0549, 0.0317]$. This estimate provides no clear evidence of either higher or lower recorded terminations. The full 2005–2021 estimate is −0.0319 (SE 0.0160; $p = 0.0497$); excluding only 2016–2017 gives −0.0309 (SE 0.0164; $p = 0.063$). These longer-window estimates are sensitivity checks. Omitting the two intervention years retains observations from both sides of the administrative change and does not establish their comparability. The difference between the estimates does not identify the effect of the administrative intervention.

Administrative registration activity gives −0.0144 (SE 0.0073; $p = 0.053$) over 2005–2021, alongside an enterprise-stock estimate of −0.0132 ($p = 0.045$). On the common 2005–2015 window, registration activity gives 0.0066 (SE 0.0087; $p = 0.447$) and enterprise stock gives −0.0180 (SE 0.0083; $p = 0.032$). The negative registration association therefore does not hold across both windows. The SII archive also publishes identical data rows in files labeled *primera inscripción* and *actividades inscritas*, although SII defines these concepts separately. The analysis reads the former file and interprets it as administrative registration activity, rather than a clean count of firm births.

Table B.8: Enterprise stocks, administrative registration activity, and terminations

Outcome	Window	Coefficient	SE	p	N
Enterprise stock	2005–2021	-0.0132**	0.0065	0.045	1,768
Registration activity	2005–2021	-0.0144*	0.0073	0.053	1,768
Current terminations	2005–2015	-0.0116	0.0218	0.595	1,144
Terminations, full window	2005–2021	-0.0319**	0.0160	0.0497	1,768
Terminations, omit 2016–2017	2005–2021	-0.0309*	0.0164	0.063	1,560

Notes: All rows use the main area-level specification and 104 areas, with area-clustered standard errors. Enterprise stock is measured in logs; registration activity and current terminations use $\log(1 + x)$. The reference termination window is 2005–2015, before SII administrative actions increased terminations in 2016–2017. The last two rows are sensitivity checks; the final row excludes 2016 and 2017. Terminations include zero-count area-years. Current terminations exclude taxpayers that restarted before extraction. Registration activity is the series published as first registrations, subject to the source duplication described above. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Fewer registrations and fewer terminations would move enterprise stock in opposite directions under a clean accounting identity. These administrative series have different coverage from the enterprise stock, and coefficients on their log transformations cannot be subtracted to decompose its change. Restricting the window also leaves the retrospective exclusion of restarters unchanged. The estimates do not identify the contribution of entry and exit to changes in enterprise stocks.

B.9 Transport-related conflict and adjustment margins

The enterprise-stock and registration associations are more negative for conflict at transport locations than for conflict at other locations. I classify attacks, coercive actions, and land invasions coded by MACEDA as ROAD, BRIDGE, or HIGHWAY as transport events; all remaining events in these three families are nontransport events. Each event must occur outside the receiving urban

area and within 25 km of its footprint, measured by exact event-point-to-footprint distance. The log of one plus each annual count enters the preferred area-level specification jointly. Location type is strongly associated with event family, so I also compare transport and nontransport locations within attacks. The transport-minus-nontransport difference in the SII series published as first registrations is nearly unchanged in that comparison, with lower precision. Enterprise stock follows the same negative ordering. The transport–nontransport differences for recorded terminations are negative but imprecisely estimated in both the full window and the pre-2016 sample.

Table B.9: Transport-related conflict and administrative adjustment margins

Outcome / specification	Transport	Nontransport	Transport − nontransport	Equality p-value
Enterprise stock	-0.0266**	-0.0081	-0.0185*	0.077
	(0.0105)	(0.0057)	(0.0104)	
SII first-registration series	-0.0400***	-0.0047	-0.0353**	0.028
	(0.0120)	(0.0076)	(0.0158)	
First-registration series: attacks only	-0.0394***	-0.0039	-0.0355*	0.078
	(0.0134)	(0.0105)	(0.0199)	
First-registration series: episode counts	-0.0341**	-0.0050	-0.0290	0.109
	(0.0133)	(0.0085)	(0.0179)	
First-registration series: attack episode counts	-0.0385**	-0.0049	-0.0336	0.122
	(0.0149)	(0.0109)	(0.0215)	
Terminations, 2005–2015	-0.0803	0.0038	-0.0841	0.160
	(0.0527)	(0.0218)	(0.0595)	
Terminations: full-window sensitivity	-0.0643**	-0.0189	-0.0455	0.283
	(0.0318)	(0.0177)	(0.0421)	

Notes: Transport events are main-family MACEDA events coded ROAD, BRIDGE, or HIGHWAY; nontransport events are the remaining main-family events. External counts use exact distance of at most 25 km from the receiving area and exclude events inside it. The two log(1+count) exposures enter jointly. All models include area and region-by-year fixed effects, three preferred predetermined trends, and contemporaneous log(1+within-area main-family events); the within-area event-count control is retained in episode specifications. Standard errors are clustered by the 104 fixed urban areas. Enterprise stock is in logs; the SII first-registration and termination series are in log(1+x). The 2005–2021 sample has 1,768 area-years; the pre-2016 termination sample has 1,144. Differences are formal linear contrasts, with equality p-values. Episode counts assign transport class if any eligible main-family record in the full episode has a transport location (attacks analogously) and count distinct episodes per receiving area-year without moving their dates. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Counting related MACEDA records as episodes preserves most of the negative registration difference, but the transport–nontransport contrasts are less precise. An episode is classified as transport if any eligible main-family record in the full MACEDA file has a transport location; the attack comparison applies the same rule to attack records. Exposure counts each distinct episode once per receiving area-year when at least one qualifying record lies outside that area and within 25 km. An episode can therefore contribute in more than one year, and timing remains tied to qualifying records rather than the episode root.

Transport and nontransport event–area pairs have similar distances: their mean and median

distances are 13.11 and 13.37 km for transport events, against 13.58 and 13.78 km for other events. Attack-only distances are also similar. With the first-registration series as the outcome, a model entering transport and nontransport conflict exposure jointly in three fixed bands gives negative transport-minus-nontransport differences at 0–10 and $>$ 10–20 km, but not at $>$ 20–25 km. The three equality restrictions reject jointly ($F(3, 103) = 3.557$, $p = 0.0169$). This pattern does not identify an optimal radius or a causal distance threshold.

The registration-source limitation and the pre-2016 reference window for terminations follow Appendix B.8. The full-window termination comparison is retained as a sensitivity check.

C Timing and conflict escalations

C.1 Continuous lead-lag results

The continuous projection in Section 4 estimates associations with future, current and past annual exposure. Figure C.1 complements Figure 3 by showing the profiles for lights, mean appraisal and newly built area alongside the three main outcomes.

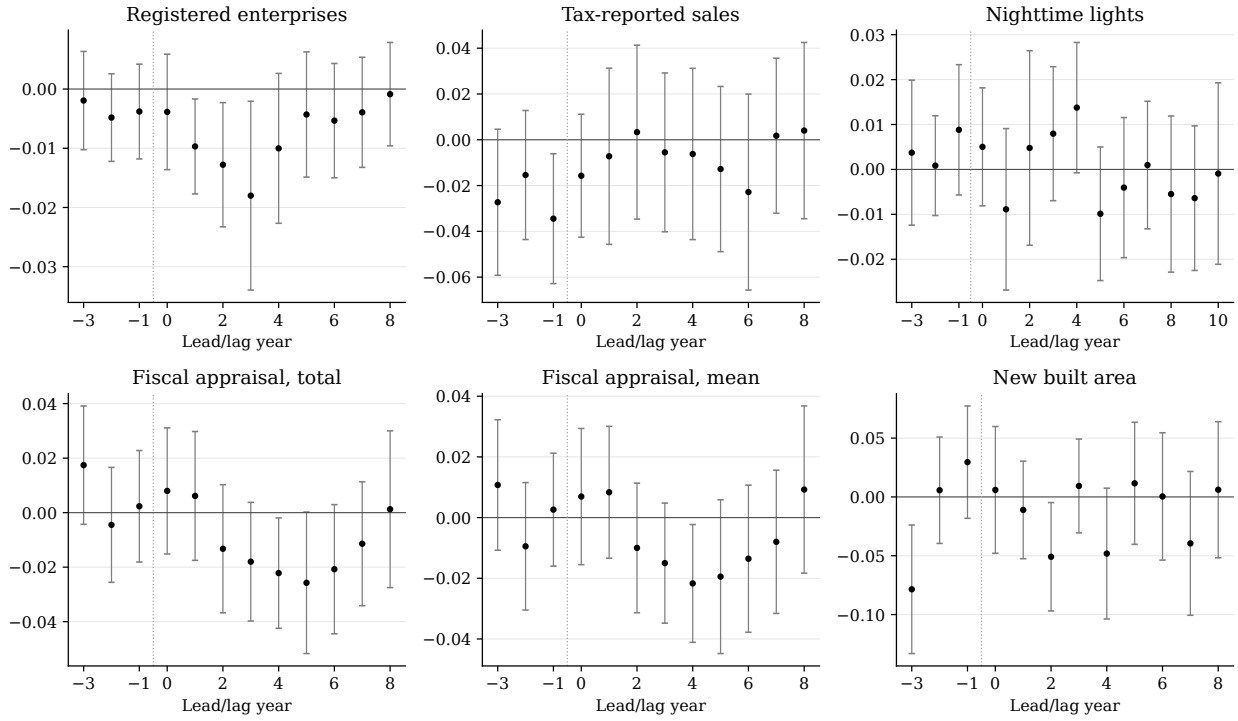


Figure C.1: Continuous lead-lag profiles for urban-area outcomes. Negative horizons denote future exposure terms, zero denotes current exposure, and positive horizons denote lagged exposure. Bars show 95 percent confidence intervals based on area-clustered standard errors.

The joint future-exposure tests in Table C.1 show the clearest pre-movement for newly built area ($p = 0.031$).

Table C.1: Summary of continuous lead-lag profiles

	Sales	Enterprises	Nighttime lights	Appraisal total	Appraisal mean	New built area
Joint leads -3/-1	2.026	0.641	0.497	1.286	1.007	3.076**
	p=0.115	p=0.590	p=0.685	p=0.283	p=0.393	p=0.031
Average current/lag 0–2	-0.007	-0.009**	0.000	0.000	0.002	-0.019
	(0.015)	(0.004)	(0.006)	(0.010)	(0.009)	(0.013)
Average lags 3–5	-0.008	-0.011*	0.004	-0.022**	-0.019**	-0.009
	(0.015)	(0.006)	(0.004)	(0.009)	(0.009)	(0.014)
Average lags 6–8	-0.006	-0.003	-0.003	-0.010	-0.004	-0.011
	(0.016)	(0.004)	(0.005)	(0.011)	(0.011)	(0.015)
Observations	1,456	1,456	1,964	936	936	1,872
Areas	104	104	104	104	104	104
Window	2005–2018	2005–2018	2000–2018	2010–2018	2010–2018	1998–2015

Notes: The joint-lead row reports the Wald statistic and its p-value for the three future-exposure coefficients. Current-and-lag rows are covariance-aware averages of the indicated current and lagged coefficients. All models use area and region-by-year effects, within-area conflict, and area-clustered SEs. Formal outcomes and appraisal use all three trends; full-window lights and built-area models omit them. Windows differ with outcome availability and lead/lag support. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Municipal timing diagnostics around conflict escalations

These comparisons align outcomes around extreme annual frequency spikes or documented destructive events. Conflict timing can respond to local conditions, so the profiles describe changes around those events.

Treatment definitions

Historical extreme-frequency spikes use the same external 25-km geography as the aggregate analysis. Let N_{mt} count eligible MACEDA attacks, coercive actions and land invasions, excluding protests. Records inside another urban area can qualify; threats and records without a specified day remain in this annual universe. Define

$$B_{mt} = \mathbf{1}\{N_{mt} \geq 3,\ N_{mt} > \mu^{10}_{mt} + 4s^{10}_{mt}\},$$
$$\mu^{10}_{mt} = \frac{1}{10}\sum_{a=1}^{10} N_{m,t-a}, \qquad (s^{10}_{mt})^2 = \frac{1}{9}\sum_{a=1}^{10}\big(N_{m,t-a} - \mu^{10}_{mt}\big)^2. \tag{4}$$

All ten preceding years must be observed. Classification starts in 2000, and earlier states are unknown. The full focal-year count determines the spike indicator. A rising threshold after an escalation can produce a subsequent zero even if conflict continues.

Documented destructive conflict events record burning or destruction of productive or freight-related assets at the same external distance. These events can occur without an unusual annual increase in frequency.

Three complementary designs

All profiles use $k = -3, -2, -1, 0, \ldots, 5$, with -1 as reference. Write $\Delta Y_{mgk} = Y_{m,g+k} - Y_{m,g-1}$. Year zero is the qualifying calendar year; $+1$ is the first complete subsequent year. Each outcome and design retains its own fixed supported population across the profile.

First-exposure C&S defines G_m as the first qualifying year in the available history and uses the absorbing experience indicator $A_{mt} = \mathbf{1}\{G_m \leq t\}$. Annual exposure can subsequently reverse. Under no anticipation and untreated parallel trends, the group-time target is $\mathbb{E}[Y_{mt}(g) - Y_{mt}(\infty) \mid G_m = g]$ (Callaway and Sant'Anna, 2021). The estimator compares mean changes within region/cohort cells, using never-exposed or not-yet-exposed donors with $G_j \neq g$ and $G_j > \max\{g-1, g+k\}$. Cells receive their observed treated-municipality shares. First experiences at or before the outcome baseline are excluded rather than moved to panel entry.

The *Repeated-Exposure Dynamic DiD Comparison* retains every supported qualifying municipality-year. Let $\mathcal{E}_m$ contain its n_m qualifying years, M the number of represented municipalities, and C_{mg} the donors. The sample contrast is

$$\widehat{\theta}_k^R = \frac{1}{M}\sum_m \frac{1}{n_m}\sum_{g\in\mathcal{E}_m}\left[\Delta Y_{mgk} - \frac{1}{|C_{mg}|}\sum_{j\in C_{mg}}\Delta Y_{jgk}\right]. \tag{5}$$

Municipalities receive equal total weight, split equally across their qualifying years; donors receive equal weight within a focal comparison. The principal same-region donors have no qualifying exposure from $g-3$ through $g+5$. A sensitivity comparison requires eligibility only through g, retaining later donor exposure on the identical focal roster and weights. Treated histories are unrestricted, so entries, continuations and re-entries enter when supported. Repeated-event and stacking methods motivate these comparisons (Cengiz et al., 2019; Schmidheiny and Siegloch, 2023; Wing et al., 2024); the particular donor rules and municipal weights define a project-specific DiD-style contrast, not an exact implementation of a named causal estimator.

Native nonabsorbing DCDH uses the first observed switch from the outcome panel's baseline dose and retains the actual subsequent annual path (de Chaisemartin and D'Haultfœuille, 2026). Baseline-zero municipalities switch into exposure; earlier historical exposure is allowed. Same-region donors remain unswitched through the paired forward horizon. The nonnormalized target compares the realized treatment path with remaining at baseline, rather than resetting the clock at every recurrence. Native effect ℓ maps to $k=\ell-1$; its backward placebo maps to $k=-1-\ell$, preserving the returned sign and covariance.

All three designs compare municipalities within regions without the main model's covariate trends or within-area-conflict control. Prior conflict histories, donor eligibility and shared incidents matter for interpretation.

Results and interpretation

Figures C.2–C.3 prioritize first and repeated exposure; Table C.2 retains all three designs. Enterprises, sales and the two reported labor margins cover 2005–2021. Lights have seven missing observations, leaving 100 complete municipalities for C&S and native inputs; repeated comparisons require complete focal windows. Fiscal appraisal covers 2010–2021 and has three supported cohorts; WSF covers 2005–2015 and has only two early cohorts. Appraisal and WSF have no common three-preyear/three-postyear calendar. SII quantities follow tax or headquarters addresses, and reported workers count employer relationships.

Frequency-spike enterprise contrasts are more negative after first experience than under repeated exposure or native switching; sales and appraisal estimates are imprecise. Lights show adverse preperiod diagnostics for repeated exposure and native DCDH ($p = 0.0063$ and 0.0046). Negative worker and WSF contrasts concern their stated populations and cannot resolve endogenous timing or shared shocks. Nonrejection of other preperiod tests does not establish parallel trends.

Documented destructive events give a different repeated-exposure profile. On its unchanged 174 qualifying years in 47 municipalities, changing from future-unexposed to initially eligible donors moves the +5 enterprise contrast from −0.0584 to −0.0280, and sales from −0.1762 to −0.0860. Continuations carry 36.0 percent of weight and contribute 54.4 percent of the net negative enterprise contrast; this is accounting, not a causal continuation effect. The frequency-spike recurrent population instead has 19 first positives, 11 re-entries and no continuations. Completed entry and cessation comparisons remain imprecise and historically imbalanced, and destructive-event C&S sales has adverse leads ($p = 0.0397$). These findings limit a persistence interpretation.

Custom covariance accumulates all treated and donor contributions within original municipality identity, retaining reused donors and overlapping windows; DCDH keeps its native covariance. Shared MACEDA incidents link municipalities, however, and full campaign-level inference is unavailable. The largest linked component covers 9 of 29 frequency-spike recurrent municipalities, versus 34 of 47 under destructive events.

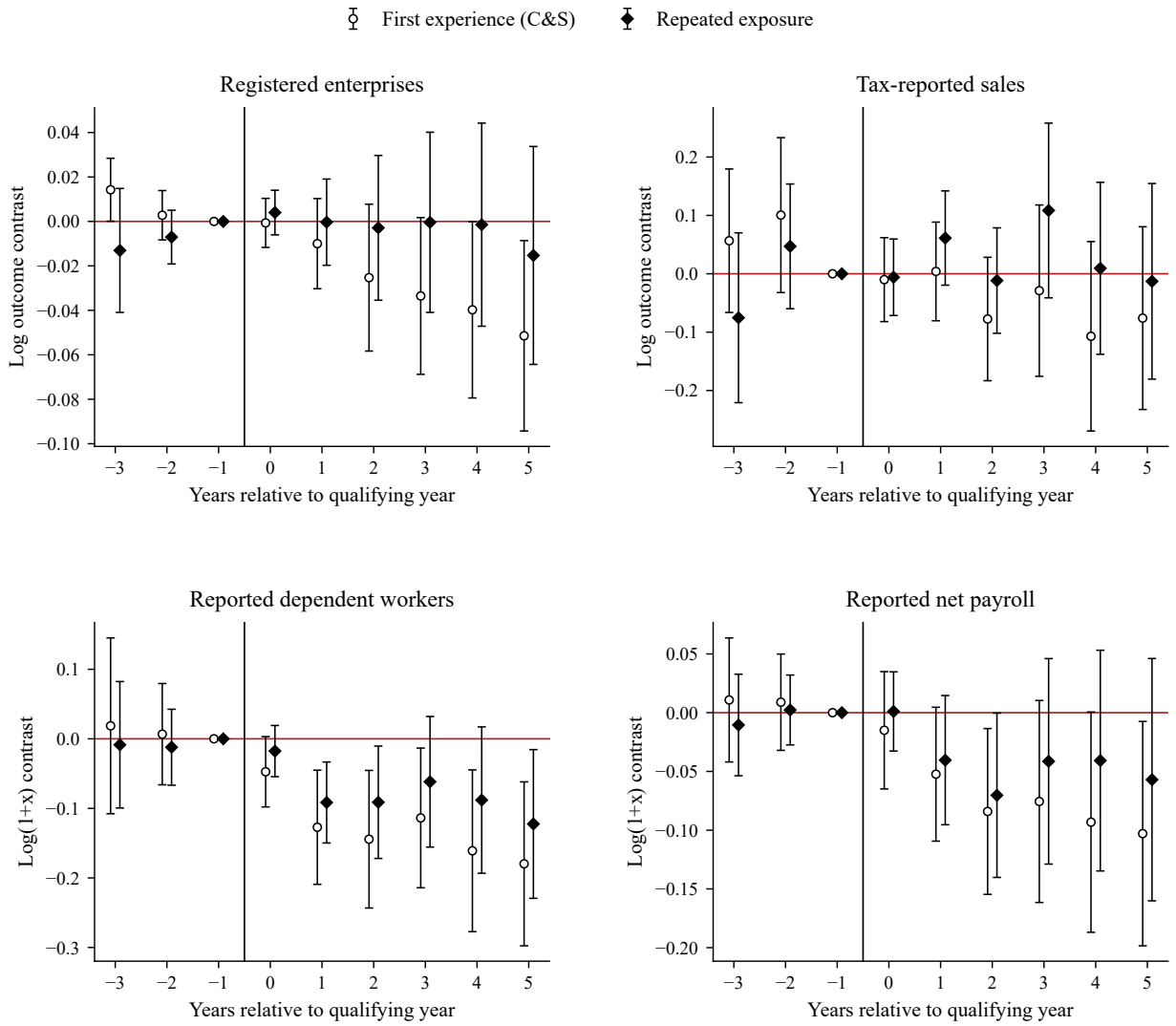


Figure C.2: First and repeated exposure around historical frequency spikes: formal activity and reported labor. Points and vertical whiskers show estimates and pointwise 95 percent municipality-conditional intervals. Open circles denote C&S; filled diamonds denote repeated exposure. Markers are offset by ± 0.09 years for visibility. The red line is zero and the black line separates -1 from zero. The omitted reference is normalized. Population and transformations are in Table C.2; DCDH is retained there.

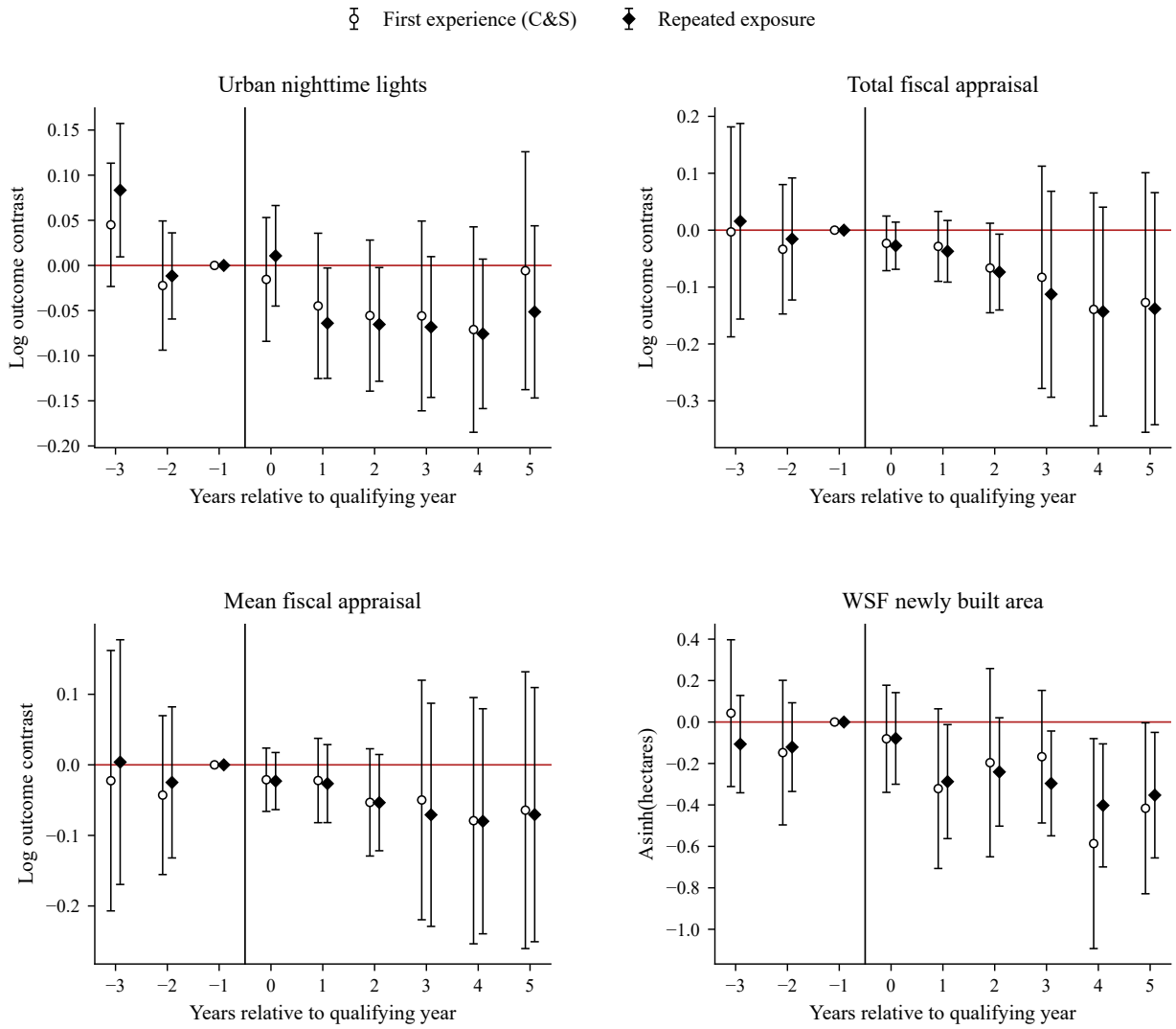


Figure C.3: First and repeated exposure around historical frequency spikes: lights, fiscal appraisal and newly built area. The figure uses the same point, whisker, offset and cutoff conventions as Figure C.2. Each curve retains its own supported −3 through +5 population. Fiscal profiles have three cohorts and WSF two early cohorts; these outcomes do not share a common event window. Preperiod tests and all native DCDH endpoints appear in Table C.2.

Table C.2: Municipal timing diagnostics: endpoints, populations and preperiods

Outcome	Estimator	N/E	$k = +3$	$k = +5$	Preperiod p
A. Historical extreme-frequency spikes					
Registered enterprises	C&S	19/19	-0.0335 (0.0180)	-0.0515 (0.0219)	0.0907
	Repeated	29/30	-0.0004 (0.0207)	-0.0153 (0.0250)	0.5147
	DCDH	29/29	0.0001 (0.0212)	-0.0169 (0.0267)	0.5531
Tax-reported sales	C&S	19/19	-0.0289 (0.0749)	-0.0760 (0.0799)	0.3213
	Repeated	29/30	0.1084 (0.0764)	-0.0129 (0.0855)	0.1585
	DCDH	29/29	0.0870 (0.0780)	-0.0369 (0.0904)	0.2043
Reported dependent workers	C&S	19/19	-0.1137 (0.0512)	-0.1797 (0.0601)	0.9578
	Repeated	29/30	-0.0618 (0.0479)	-0.1224 (0.0545)	0.9099
	DCDH	29/29	-0.0679 (0.0509)	-0.1330 (0.0599)	0.9465
Reported net payroll	C&S	19/19	-0.0756 (0.0439)	-0.1029 (0.0487)	0.9048
	Repeated	29/30	-0.0414 (0.0446)	-0.0570 (0.0526)	0.7328
	DCDH	29/29	-0.0483 (0.0467)	-0.0668 (0.0557)	0.8389
Urban nighttime lights	C&S	19/19	-0.0559 (0.0536)	-0.0058 (0.0672)	0.1486
	Repeated	29/30	-0.0682 (0.0398)	-0.0515 (0.0487)	0.0063
	DCDH	29/29	-0.0662 (0.0441)	-0.0452 (0.0496)	0.0046
Total fiscal appraisal	C&S	10/10	-0.0829 (0.0997)	-0.1271 (0.1164)	0.4623
	Repeated	11/11	-0.1126 (0.0923)	-0.1380 (0.1041)	0.6290
	DCDH	11/11	-0.0941 (0.0742)	-0.1380 (0.0838)	0.5543
Mean fiscal appraisal	C&S	10/10	-0.0498 (0.0866)	-0.0642 (0.1001)	0.4655
	Repeated	11/11	-0.0708 (0.0807)	-0.0706 (0.0919)	0.5400
	DCDH	11/11	-0.0580 (0.0744)	-0.0706 (0.0850)	0.4293
WSF newly built area	C&S	9/9	-0.1673 (0.1632)	-0.4162 (0.2104)	0.5427
	Repeated	19/19	-0.2963 (0.1290)	-0.3529 (0.1545)	0.5014
	DCDH	19/19	-0.2963 (0.1319)	-0.3529 (0.1597)	0.4850
B. Documented destructive conflict events					
Registered enterprises	C&S	30/30	0.0053 (0.0191)	-0.0262 (0.0276)	0.2518
	Repeated	47/174	-0.0240 (0.0164)	-0.0584 (0.0234)	0.6479
	DCDH	36/36	0.0043 (0.0167)	-0.0275 (0.0245)	0.3238
Tax-reported sales	C&S	30/30	-0.0176 (0.0830)	-0.0560 (0.0993)	0.0397
	Repeated	47/174	-0.1527 (0.1141)	-0.1762 (0.0835)	0.1943
	DCDH	36/36	0.0027 (0.0741)	-0.0427 (0.0927)	0.0952

Notes: Standard errors in parentheses. N/E is treated municipalities / focal years; first-experience and first-switch designs have one focal year per municipality. Both endpoints use the same supported -3 through $+5$ profile. Enterprises, sales, lights and appraisal are logs; workers and net payroll use $\log(1+x)$, and WSF uses asinh hectares. Payroll is net remuneration in UF, not firm profits. Baselines are 2005, except fiscal 2010. Frequency-spike formal/lights profiles have five cohorts, fiscal three and WSF two. C&S and repeated preperiod columns test the two nontrivial leads; native DCDH uses its two backward-placebo joint test. Tests are unadjusted. The research archive retains initial-eligibility estimates and the simultaneous post-period families (48 C&S cells, 96 across both repeated rules, and 48 native Bonferroni cells for frequency spikes), alongside the destructive-event families.

D Within-area spatial incidence

The aggregate estimates describe how formal activity varies across urban areas over time. Table D.1 instead compares fixed 1-kilometer grid cells within the same area-year. At a given radius R, an external event at distance d from a grid-cell centroid contributes $\max\{0, 1 - d/R\}$ to S_R. Thus, an event at the cell receives weight one, an event halfway to the stated radius receives weight one-half, and an event at or beyond the radius receives zero. S_R sums these weighted attacks, coercive actions, and land invasions outside all urban systems, and the regression uses $\log(1 + S_R)$. Each specification also includes direct local main conflict, $\log(1 + L)$, inside the same grid cell.

Grid-cell fixed effects absorb persistent differences across locations, and area-by-year fixed effects absorb shocks shared within an urban area. The selected legal-entity count is more negative at 5 and 10 kilometers, but is close to zero or positive at 20, 25, and 30 kilometers. Commercial-use share is negative at 10, 20, 25, and 30 kilometers, with a similarly negative point estimate at 5 kilometers. Nighttime lights, fiscal appraisal, and new built area do not show a common negative spatial gradient. This local within-area spatial-incidence profile does not identify aggregate urban decline, changes in the location of activity, or a causal radius. Legal-entity locations come from selected headquarters records and cannot be read as all firms or as locations of production or sales.

Table D.1: Within-area spatial incidence across external radii

	Selected legal entities	Urban nighttime lights	Fiscal appraisal total	Fiscal appraisal mean	Commerce-use share	New built area
5 km	-0.1141***	0.0356	0.0497	0.0217	-0.1012*	-0.0010
	(0.0275)	(0.0475)	(0.0902)	(0.0637)	(0.0561)	(0.0282)
10 km	-0.0564**	-0.0057	0.0539	0.0190	-0.1578***	0.0467
	(0.0272)	(0.0582)	(0.0704)	(0.0594)	(0.0522)	(0.0322)
20 km	0.0419	-0.0878	0.1051	0.0112	-0.1677**	0.0154
	(0.0465)	(0.0836)	(0.0979)	(0.0508)	(0.0717)	(0.0574)
25 km (preferred)	0.0428	-0.0659	0.1332	0.0305	-0.1618**	0.0161
	(0.0648)	(0.0805)	(0.0920)	(0.0526)	(0.0629)	(0.0440)
30 km	0.0446	-0.0452	0.1536*	0.0606	-0.1637***	0.0230
	(0.0783)	(0.0779)	(0.0851)	(0.0507)	(0.0614)	(0.0364)
Observations	27,591	27,591	19,014	19,014	15,901	17,853
Fixed 1-km grid cells	1,623	1,623	1,623	1,623	1,623	1,623
Areas	104	104	104	104	104	104
Outcome years	2005–2021	2005–2021	2010–2021	2010–2021	2012–2021	2005–2015

Notes: Each column includes fixed effects for 1-kilometer grid cells and area-by-year fixed effects. For radius R, the R-only external exposure assigns an eligible event at distance d a weight $\max\{0, 1 - d/R\}$: one at the grid cell, one-half halfway to R, and zero at or beyond R. $\log(1 + S_R)$ sums these weighted attacks, coercive actions, and land invasions outside all urban systems. Direct local main conflict, $\log(1 + L)$, inside the same grid cell is included in every specification. Standard errors are clustered by fixed urban area. Coefficients and standard errors are reported to four decimal places. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Municipal-versus-urban interpretation

E.1 Registered legal entities

The SII outcomes used in the main analysis are assigned through municipal tax addresses and therefore do not mechanically locate economic activity inside the fixed urban footprints. I examine

this measurement issue using the registered legal-entity (*persona jurídica*) panel. Its total count includes legal entities before temporal assignment and cadastral classification. The urban and rural subsets retain entities whose historical headquarters can be matched with high confidence to cadastral parcels.

The urban-headquarters coefficient is −0.0466 (SE 0.0221); rural headquarters give 0.0073 (SE 0.0269). Zero-count area-years are retained. The uncertain rural result does not establish an urban-versus-rural difference or an exclusively urban association. The samples are selected, and headquarters need not identify production or sales locations.

Table E.1: External conflict and registered legal entities

Outcome	Transformation	Coefficient	Clustered SE	Zero area-years
Registered enterprises	$\log(x)$	-0.0132**	0.0065	0
Total registered legal entities	$\log(x)$	-0.0073	0.0107	0
Cadastre-matched legal entities	$\log(1+x)$	-0.0335	0.0230	109
Cadastre urban legal entities	$\log(1+x)$	-0.0466**	0.0221	152
Cadastre rural legal entities	$\log(1+x)$	0.0073	0.0269	913

Notes: The total legal-entity count precedes temporal assignment, cadastral matching, and classification. The cadastral-matched sample is the parent for the urban and rural rows. Urban and rural subsets use historical headquarters matched to cadastral parcels through administrative address records. These linked subsets are selected, and headquarters locations need not identify production or sales locations. Zero-count area-years are retained. Standard errors are clustered by area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.2 Rural production linkages

A second concern is that municipality-level SII outcomes could primarily reflect businesses tied to the surrounding rural economy rather than activity associated with urban areas. I examine this possibility using predetermined sectoral linkages from the Chilean input-output matrices (*Matriz Insumo-Producto*, MIP) published by the Banco Central de Chile.

For each sector, I construct two measures of its direct connection to rural production. The rural input linkage is the share of the sector's domestic intermediate inputs purchased from agriculture, fruit growing, livestock, and forestry. The rural customer linkage is the share of the sector's domestic intermediate sales sold to those same rural activities. Fishing and aquaculture are excluded from the rural group. The 2003 MIP is the preferred benchmark because it is the closest pre-period matrix to the 2005–2021 estimation window; the 1996 MIP provides an earlier benchmark. The linkage measures are standardized across the ten comparable non-rural sectors retained in the preferred sector panel.

Let L_s denote either standardized linkage for sector s. I estimate

$$Y_{mst} = \beta\,(D_{mt} \times L_s) + \rho\,(I_{mt} \times L_s) + \alpha_{ms} + \gamma_{st} + \delta_{mt} + \varepsilon_{mst}, \tag{6}$$

where D_{mt} is external conflict exposure and I_{mt} is within-area conflict. Area-by-sector fixed effects absorb persistent differences in local sector composition, sector-by-year fixed effects absorb

national sectoral shocks, and area-by-year fixed effects absorb all shocks common to sectors in the same area and year. The coefficient β therefore asks whether the formal-activity association is systematically different in sectors that are more structurally connected to rural production.

Table E.2: External conflict and predetermined rural sectoral linkages

Benchmark	Linkage	Sales interaction	Enterprise interaction
2003	Customer	-0.0168 (0.0279)	0.0046 (0.0050)
2003	Input	-0.0008 (0.0382)	-0.0034 (0.0034)
1996	Customer	-0.0167 (0.0378)	0.0021 (0.0054)
1996	Input	0.0033 (0.0369)	-0.0028 (0.0036)

Notes: Input and customer linkages are constructed from the Banco Central de Chile input-output matrices and standardized across the ten sectors in the preferred sample. The 2003 matrix is the main pre-period benchmark and the 1996 matrix provides an earlier benchmark. Each coefficient is the interaction between log external conflict exposure within 25 kilometers and the indicated standardized sectoral linkage. Specifications also include the corresponding interaction with within-area conflict and absorb area-by-sector, sector-by-year, and area-by-year fixed effects. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction estimates are small and do not display a systematic negative gradient for either sales or enterprises. The conclusion is similar using the 2003 input and customer linkages and the earlier 1996 benchmark. The aggregate formal-activity association therefore does not appear to be disproportionately concentrated in sectors with stronger structural links to rural production.

F Historical heterogeneity

I measure historical claims by the share of terrestrial land in the receiving area's external 0–25 km catchment covered by *Títulos de Merced* (TDM). The 1992 urban Mapuche population share is reconstructed from individual census records and harmonized to the fixed urban geography. TDM coverage is available for all 104 areas. The Mapuche-share and joint models use 102 areas because Curaco de Vélez and Palena have no urban population records in the 1992 census geography.

The separate and joint estimates show no systematic amplification of the formal-business association by either moderator. TDM coverage is concentrated: 43 areas have no titled land in the external 25 km catchment, and much of the positive variation is in Araucanía. Varying the TDM catchment from 10 to 50 km while holding conflict exposure at 25 km yields no consistent gradient. A separate 30 km specification that varies both boundaries and allows conflict slopes to differ by region gives the same qualitative conclusion. These comparisons leave open how property-rights uncertainty and perceptions affect firms' decisions.

Table F.1: Municipal conflict associations by historical land coverage and population

	Separate models		Joint model (102 areas)	
Outcome	TDM (104 areas)	Mapuche (102 areas)	TDM	Mapuche
Registered enterprises	-0.009 (0.010)	0.010 (0.013)	-.009 (.011)	.010 (.015)
Tax-reported sales	0.000 (0.015)	0.005 (0.029)	-.001 (.021)	-.004 (.033)
Urban nighttime lights	-0.011 (0.008)	-0.005 (0.015)	-.013 (.009)	.007 (.015)
Fiscal appraisal total	-0.007 (0.013)	0.025 (0.021)	-.019 (.017)	.036 (.024)
Fiscal appraisal mean	-0.003 (0.012)	0.013 (0.019)	-.009 (.015)	.019 (.022)
WSF newly built area	-0.009 (0.031)	-0.030 (0.037)	.006 (.029)	-.033 (.038)

Notes: External-exposure interaction coefficients per standard deviation of the moderator, with area-clustered standard errors in parentheses. Models include area and region-by-year effects, within-area conflict, and the corresponding moderator trends and interactions with within-area conflict. Formal outcomes and lights use 2005–2021, and appraisal uses 2010–2021, with the three predetermined trends. Built area omits those trends and covers 1993–2015 in the separate models and 1992–2015 in the retained joint estimates. The original joint wild-cluster tests all give $p \geq 0.174$.

G Institutional response: CONADI

The published CONADI GIS layers cover land-acquisition subsidies under Article 20(a) and land claims under Article 20(b). I pool the records and assign them by parcel municipality. Polygon parts with a common acquisition record key count as one group, and reported hectares sum across parts. The GIS year does not distinguish administrative stages such as approval, registration, and delivery.

Greater external exposure precedes more recorded acquisition groups and reported hectares at the two-year lag. The one-year estimates are imprecise, and the sum of the current and two lagged coefficients is positive for both outcomes. Acquisitions may respond to conflict or to persistent claims that also influence conflict. The estimates describe recorded acquisition timing and leave the effects of restitution on business activity and subsequent violence unresolved.

Table G.1: External conflict exposure and recorded CONADI land acquisition

Exposure term	Recorded acquisition groups	Reported hectares
Current exposure (static)	0.0751** (0.0320)	0.1707* (0.0873)
Current exposure (distributed lag)	0.0675* (0.0343)	0.1582* (0.0935)
One-year lag	-0.0051 (0.0295)	-0.0525 (0.0923)
Two-year lag	0.0896*** (0.0263)	0.2456*** (0.0871)
Joint p-value	0.003	0.022
Sum of current and lagged coefficients	0.1519*** (0.0442)	0.3513** (0.1378)

Notes: The static model uses current exposure. The distributed-lag model includes current exposure and two annual lags. The sum uses their full covariance matrix, and the joint test covers all three terms. Models use the preferred 104-area specification for 2005–2021, with area-clustered standard errors in parentheses. Outcomes are transformed as $\log(1+x)$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.